

RESEARCH ARTICLE

DiabetesXpertNet: An innovative attention-based CNN for accurate type 2 diabetes prediction

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Abstract

Type 2 diabetes mellitus remains a critical global health challenge, with rising incidence rates placing immense pressure on healthcare systems worldwide. This chronic metabolic disorder affects diverse populations, including the elderly and children, leading to severe complications. Early and accurate prediction is essential to mitigate these consequences, yet traditional models often struggle with challenges such as imbalanced datasets, high-dimensional data, missing values, and outliers, resulting in limited predictive performance and interpretability. This study introduces DiabetesXpertNet, an innovative deep learning framework designed to enhance the prediction of Type 2 diabetes mellitus. Unlike existing convolutional neural network models optimized for image data, which focus on generalized attention mechanisms, DiabetesXpertNet is specifically tailored for tabular medical data. It incorporates a convolutional neural network architecture with dynamic channel attention modules to prioritize clinically significant features, such as glucose and insulin levels, and a context-aware feature enhancer to capture complex sequential relationships within structured datasets. The model employs advanced preprocessing techniques, including mean imputation for missing values, median replacement for outliers, and feature selection through mutual information and LASSO regression, to improve dataset quality and computational efficiency. Additionally, a logistic regression-based class weighting strategy addresses class imbalance, enhancing model fairness. Evaluated on the PID dataset and Frankfurt Hospital, Germany Diabetes datasets, DiabetesXpertNet achieves an accuracy of 89.98%, AUC of 91.95%, precision of 89.08%, recall of 88.11%, and F1-score of 88.01%, outperforming existing machine learning and deep learning models. Compared to traditional machine learning approaches, it demonstrates significant improvements in precision (+5.1%), recall (+4.8%), F1-score (+5.1%), accuracy (+6.0%), and AUC (+4.5%). Against other convolutional neural network models, it shows meaningful gains in precision (+2.2%), recall (+1.1%), F1-score (+1.2%), accuracy (+1.9%), and AUC (+0.6%). These results underscore the robustness and interpretability of DiabetesXpertNet, making it a promising tool for early Type 2 diabetes diagnosis in clinical settings.

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Data availability statement: The Pima Indians Diabetes Dataset (PIDD) can be accessed on the Kaggle website: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>. The Frankfurt Hospital, Germany Diabetes Dataset (FHGDD) is available on GitHub at: https://github.com/ibrahimuhammad/diabetes_prediction/blob/main/diabetes.csv

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1. Introduction

Diabetes mellitus, one of the most widespread and significant chronic diseases worldwide, is caused by abnormally high blood glucose levels [1,2]. This condition severely impacts individuals' quality of life and overall health, posing a substantial challenge to healthcare systems in both developed and developing countries [3]. The International Diabetes Federation estimates that 451 million people worldwide had diabetes in 2017, and by 2045, there would be 693 million people with this disease. This dramatic increase, particularly in developing countries, underscores the critical importance of addressing this disease [4,5]. The three main forms of diabetes are type 1 diabetes mellitus (T1DM), type 2 diabetes mellitus (T2DM), and gestational diabetes mellitus (GDM) [6]. T1DM typically manifests in individuals under the age of 30 and requires careful management and insulin therapy [7]. GDM is a common complication during pregnancy and can have severe consequences for both the mother and the fetus [8].

T2DM, which accounts for 90–95% of diabetes cases [9,10], predominantly affects middle-aged and elderly individuals and is often associated with obesity, hypertension, and cardiovascular diseases [11,12]. This disease is complex and progressive, posing significant challenges to patients and healthcare systems worldwide [12]. Unlike T1DM, T2DM often develops insidiously over many years, with early symptoms that are subtle or even absent [12]. As a result, many individuals with T2DM are undiagnosed until they experience severe complications [13]. These complications, including retinopathy, nephropathy, neuropathy, and cardiovascular disease, not only increase morbidity and mortality rates but also lead to a significant reduction in quality of life for patients [14,15].

The financial burden related to T2DM is substantial. It includes direct medical expenses, including hospitalizations, medications, and treatments, and indirect expenses, for example, lost productivity and disability [16,17]. In many countries, the increasing prevalence of T2DM has strained medical resources, causing more efficient strategies for early detection and intervention [13,18]. Early and quick diagnosis of T2DM is essential because it has the potential to avoid or delay the onset of these crippling complications, which decreases the total cost to patients and the medical system [19,20]. In recent years, studies have increased their focus on enhancing early diagnosis and prediction of T2DM using cutting-edge techniques like machine learning models (MLMs) and deep learning models (DLMs) [5,21]. This necessity underscores the importance of continuous exploration and innovation in this field. Refining these algorithms could lead to the development of tools that are better suited for identifying individuals at risk for T2DM in the early stages, enhancing patient outcomes and reducing the long-term expenses related to the disease.

To address these challenges, we propose DiabetesXpertNet, a novel Convolutional Neural Network (CNN)-based framework for T2DM prediction that overcomes limitations of existing ML and DL models, such as handling imbalanced datasets, high-dimensional data, and data quality issues (e.g., missing values, outliers). Unlike standard CNNs and attention-based CNNs like Squeeze-and-Excitation Networks

(SE-Net) [22] and Convolutional Block Attention Module (CBAM) [23], which are primarily designed for image data, DiabetesXpertNet is tailored for tabular medical data. It integrates a Dynamic Channel Attention Module (DCAM) that uses GlobalAveragePooling1D and dense layers to adaptively prioritize clinically relevant features (e.g., glucose, insulin) based on their predictive power, and a context-aware feature enhancer (C-AFE) that employs Conv1D, Lambda, and Concatenate operations to capture complex sequential and contextual relationships in structured datasets. While SE-Net focuses on channel-wise recalibration and CBAM combines channel and spatial attention for image processing, DiabetesXpertNet's design emphasizes sequential patterns in tabular data, avoiding the need for data transformation (e.g., converting numerical data to images as in [28]), thus preserving feature integrity and reducing preprocessing overhead. These innovations offer superior interpretability and efficiency (~1.2M parameters vs. ~1.5M for CBAM-based CNNs), distinguishing DiabetesXpertNet from traditional and attention-based models. Evaluated on the PIMA Indian Diabetes (PID) Dataset and Frankfurt Hospital, Germany Diabetes (FHGD) Dataset, DiabetesXpertNet demonstrates robust performance, advancing AI-driven early T2DM diagnosis. The contributions of this study are as follows:

- (1) Bridging Gaps in Existing Models: DiabetesXpertNet addresses the limitations inherent in traditional MLMs and DLMs. Unlike support vector machines (SVM) and random forests (RF), which struggle with high-dimensional datasets, and attention-based CNNs like SE-Net [22] and CBAM [23], which are optimized for image data, our model is tailored for tabular medical data. By integrating a DCAM and a C-AFE, DiabetesXpertNet demonstrates superior performance over standard 1D CNNs in identifying critical features and capturing sequential patterns, which is crucial for accurate diabetes prediction.
- (2) Enhancing Dataset Quality: To improve the reliability and robustness of the dataset, comprehensive preprocessing techniques were employed:
 - Median Imputation: This method addresses missing data while maintaining key insights, ensuring the dataset's integrity.
 - Outlier Management: The model handles outliers by retaining valuable but unconventional data points, ensuring algorithm stability.
 - Feature Selection via mutual information and LASSO Regression: By reducing redundancy, this approach enhances computational efficiency, interpretability, and accuracy of the model.
- (3) Class Imbalance Management: To address class imbalance and enhance model fairness, a refined class weighting approach with Logistic Regression (LR) was employed. A detailed range of weights for the minority class, ranging from 0 to 1 in increments of 0.0005, was explored, utilizing GridSearchCV for meticulous hyperparameter optimization. The model's performance was rigorously evaluated through 10-fold cross-validation with StratifiedKFold to ensure reliability and robustness. The optimal class weights were selected based on their impact on model accuracy and were fine-tuned to four decimal places. This strategic adjustment allowed the LR model to manage class imbalances effectively, leading to improved predictive accuracy across all classes. By integrating this class weighting strategy, the adverse effects of class imbalance were successfully mitigated, achieving a more balanced and effective model.
- (4) Introducing an Innovative Model Design: DiabetesXpertNet introduces a cutting-edge architecture that is optimized to tackle common challenges:
 - DCAM: This module enhances feature prioritization by adaptively weighting key attributes, improving the focus on the most relevant features.
 - C-AFE: The use of Conv1D layers refines both local and spatial data representations, leading to more precise pattern recognition.

- Regularization and Dropout Layers: These components mitigate overfitting, ensuring improved generalizability and robustness across diverse datasets.
- (5) Enhancing Model Generalizability with Dual-Dataset Validation: To enhance the generalizability of the proposed model, we initially evaluated its performance on the PID dataset and subsequently incorporated the FHGD dataset for further validation. This dual-dataset approach ensures robust performance across diverse populations, addressing variability in medical data characteristics.
- (6) Rigorous Cross-Validation Techniques: Robust validation protocols were implemented to ensure the model's reliability and generalizability:
- 3-Fold Cross-Validation: This technique was used to fine-tune hyperparameters for optimal performance.
 - 10-Fold Cross-Validation: Employed to evaluate the final model's accuracy and robustness across varied data distributions, ensuring the model's effectiveness across different subsets of data.
- (7) Comprehensive Evaluation and Results: DiabetesXpertNet demonstrated exceptional accuracy and capability in identifying complex medical patterns and predicting T2DM across two datasets: PID and FHGD datasets. Compared to previous approaches and current CNN-based methods, DiabetesXpertNet showed significant advancements across all key evaluation metrics on both datasets.

These results highlight the robustness of DiabetesXpertNet in addressing key challenges in diabetes prediction, with superior performance over traditional MLM and competitive improvements over CNN-based DLMs across both PID and FHGD datasets.

2. Related work

This section explores recent advancements in diabetes prediction by reviewing the literature from two critical perspectives: data preprocessing techniques and model classification methods. This overview highlights contemporary studies, reflecting the latest improvements in the field and providing a context for understanding this study's contributions.

Recent advancements in diabetes prediction have emphasized improved data preprocessing and model classification to increase predictive accuracy. For example, Nnamoko and Korkontzelos [24] addressed the challenges of outlier detection and class imbalance by integrating outlier information with SMOTE, enhancing classifier performance. Olisah *et al.* [25] improved diagnostic accuracy through advanced feature selection and missing value imputation on the PID Dataset. Hasan Kazi Amit and Hasan Md. Al Mehedi [26] explored various classification models and reported that the random forest (RF) model was particularly effective, with an accuracy of 90%. Further developments include Oualid *et al.* [27], who introduced an innovative classification method using an isolation forest and SMOTE T-link, achieving an area under the receiver operating characteristic curve (AUC) of 97.2% and a specificity of 91.7%. Sivaranjani *et al.* [28] employed principal component analysis (PCA) and mean imputation before classifying data with a support vector machine (SVM) and RF and reported that RF is superior to SVM, with an accuracy of 83%. Khanam and Foo [29] evaluated eight MLMs, with neural networks reaching the highest accuracy of 88.6%. Ramadhan *et al.* [30] proposed a preprocessing framework for early T2DM detection, which uses median imputation and random oversampling to improve precision and recall. Aslan Aslan and Sabanci [31] made significant advancements in early diabetes detection by transforming numerical data into image-based formats and utilizing CNNs and SVMs to enhance accuracy substantially. Bukhari *et al.* [32] used the ABP-SCGNN algorithm in an advanced artificial neural network (ANN) model and demonstrated 93% accuracy on the PID dataset. Rastogi *et al.* [33] compared the LR, SVM, RF, and naive Bayes (NB) methods on the PID dataset, with the LR method achieving the highest accuracy of 82.46%. Bhattacharjee *et al.* [34] focused on lifestyle factors such as obesity, improving ML model effectiveness through preprocessing, with the LR

model achieving 81% accuracy. Deo and Panigrahi [35] used SMOTE and PCA for feature extraction, reaching 91% accuracy with a linear SVM model.

Our proposed DiabetesXpertNet addresses limitations in these approaches by introducing a CNN-based architecture specifically designed for T2DM prediction from tabular medical data. Unlike Aslan and Sabanci [31], who convert numerical data into images for CNN processing, DiabetesXpertNet directly processes structured tabular data, avoiding the loss of feature granularity inherent in image transformation. Compared to attention-based CNNs like SE-Net [22] and CBAM [23], which are optimized for high-dimensional image data and employ generalized attention mechanisms, DiabetesXpertNet incorporates a DCAM to selectively emphasize clinically critical features (e.g., glucose, insulin) and a C-AFE to capture sequential dependencies in tabular data. This targeted design reduces model complexity (~1.2M parameters vs. ~1.5M for CBAM-based CNNs) and enhances interpretability, making it more suitable for medical diagnostics. Additionally, unlike traditional ML models (e.g., RF, SVM) that struggle with high-dimensional feature interactions, DiabetesXpertNet leverages Conv1D layers to model local and global patterns, achieving superior performance. By evaluating performance on both PID and FHGD datasets, our study fills gaps in generalizability and computational efficiency, advancing AI-driven diabetes prediction.

Table 1, provide a detailed overview of the interconnected investigations, describing the data preprocessing techniques, the applied models, and the resulting accuracies. These studies collectively emphasize the significant role of data preprocessing and innovative modeling approaches in advancing methodologies for diabetes prediction.

Table 1. Overview of Related Studies in Diabetes Prediction, Focusing on Data Preprocessing and Innovative Approaches.

Authors	Years	Preprocessing Method	Models	Accuracy (%)
Hayashi and Yukita [36]	2016	Algorithm for extracting rules, Sampling selection approach.	Rules extraction and J48 graft	83.83
Zou <i>et al.</i> [38]	2018	reduction of dimensions (PCA, mRMR)	Decision Tree (DT), RF, ANN	80.84
Gnanadass [39]	2020	Filling in missing data (mean).	AdaBoost gradient boosting machine (AdaBoost), NB, RF, LR, and extreme gradient boosting.	78.00
Kumar <i>et al.</i> [40]	2020	Removal of correlated features, normalization with MinMaxScaler, handling missing values by removing rows, replacing with the mean	MultiLayer Feed Forward Neural Networks (MLFNN).	84.17
Joshi and Dhakal [41]	2021	LR, DT.	LR, DT.	78.26
Khanam <i>et al.</i> [29]	2021	Filling in missing data (mean), Outlier removal, Feature selection, Normalization.	LR, SVM, NN, AB, K-NNRF, DT.	88.60
Sivaranjani <i>et al.</i> [28]	2021	Feature selection, Dimensionality reduction.	SVM, RF.	83.00
Butt <i>et al.</i> [42]	2021	Different models and classifiers	LSTM, MLP, and RF	86.00
Krishnamoorthi <i>et al.</i> [43]	2022	Processing missing values, eliminating outliers, and normalizing.	LR.	83.00
Saxena <i>et al.</i> [44]	2022	Feature selection, rejection of outliers, and padding of missing values	K-NN, RF.	79.80
Chang <i>et al.</i> [45]	2023	Feature selection using PCA, k-means clustering, and relevance ranking; imputation of median values; treatment of outliers.	NB, RF, J48.	79.57
Abnoosian <i>et al.</i> [46]	2023	Remove Duplicate Samples, Attribute Conversion, Handling Missing or Null Values, Normalization, Standardization.	EMLM.	97.00
Rastogi <i>et al.</i> [33]	2023	—	LR, NB, SVM, RF.	82.46

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3. Material and methods

This section outlines the materials and the methods used in the present study. It begins with an overview of the data, detailing its sources, attributes, and preprocessing steps. The following section introduces the proposed framework, emphasizing its architecture and the methodologies implemented to achieve the study's objectives. These components establish a robust foundation for understanding the study and conducting following analyses.

3.1. Dataset description

The (PID dataset, a freely accessible resource from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), is a cornerstone of diabetes research. It includes data from 768 female patients of Pima Indian descent near Phoenix, Arizona, USA, with 268 diabetes cases and 500 non-diabetic cases. Widely used to improve predictive models, PID dataset features key physiological indicators like glucose levels, BMI, and blood pressure, essential for assessing diabetes risk.

To enhance the generalizability of our analysis, we incorporated the FHGD dataset. Sourced from retrospective clinical records at FHGD dataset comprised 2000 samples (1316 non-diabetic, 684 diabetic). The dataset represents a diverse European cohort, including both male and female patients with an approximate age range of 18–80 years, providing a broader demographic scope compared to the Pima Indian-specific PID and FHGD datasets shares the same eight features as PID dataset (Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age), ensuring consistency in feature structure for comparative analysis, as outlined in [Table 2](#). The inclusion of FHGD dataset enhances the study's robustness by validating DiabetesXpertNet across diverse populations, improving its applicability in varied clinical settings.

Both datasets exhibit similar patterns in feature distributions, as shown in [Figs 1 and 2](#), reinforcing their suitability for evaluating DiabetesXpertNet. [Table 2](#) outlines the attributes of both datasets, providing a clear overview of their characteristics and distribution. [Fig 1](#) displays these attributes distribution for the PID dataset, while [Fig 2](#) shows the FHGD dataset distribution, with blue representing non-diabetic individuals (class 0) and orange representing diabetic individuals (class 1). These visual aids, combined with tabular summaries, are essential for analyzing the datasets.

3.2. Proposed framework

The proposed framework for T2DM prediction, shown in [Fig 3](#), incorporates advanced preprocessing techniques and innovative neural network architectures. Key contributions include the development of the DiabetesXpertNet model, the

Table 2. Overview of the PID and FHGD Datasets.

Attributes	Description
Pregnancies	The quantity of pregnancies the patient has experienced.
Glucose	Plasma Glucose Concentration a 2 Hours in an Oral Glucose Tolerance Test.
Blood Pressure	Blood pressure at diastole (mm.Hg).
Skin Thickness	Skin Fold Thickness (mm) of the Triceps.
Insulin	Serum Insulin (muU/ml) after 2 hours.
BMI	Body Mass Index (kg of weight divided by in-inch height) ²).
Diabetes Pedigree Function	A function that scores the likelihood of diabetes based on family history and genetic factors.
Age	Patient's age expressed in years.
Outcome	Whether or not the patient has diabetes is indicated by the class variable (0 or 1).

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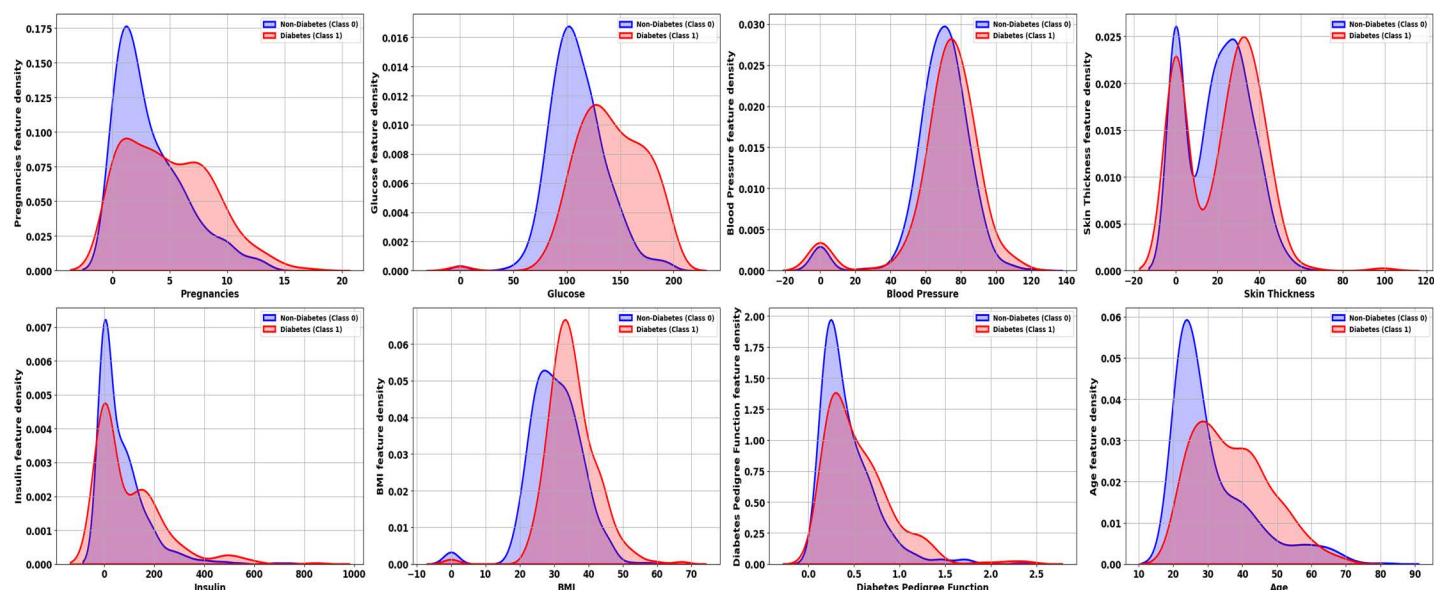


Fig 1. Population distribution of PID dataset (Non-diabetic: blue, Diabetic: orange).

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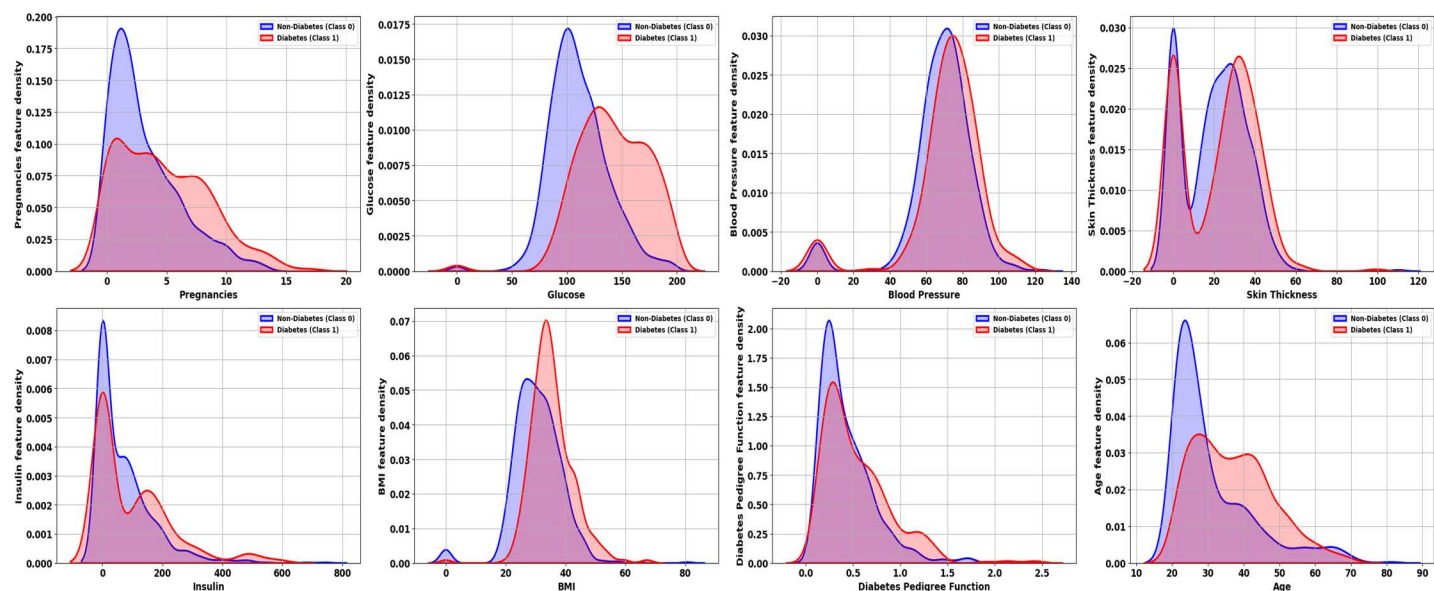


Fig 2. Population distribution of FHGD dataset (Non-diabetic: blue, Diabetic: orange).

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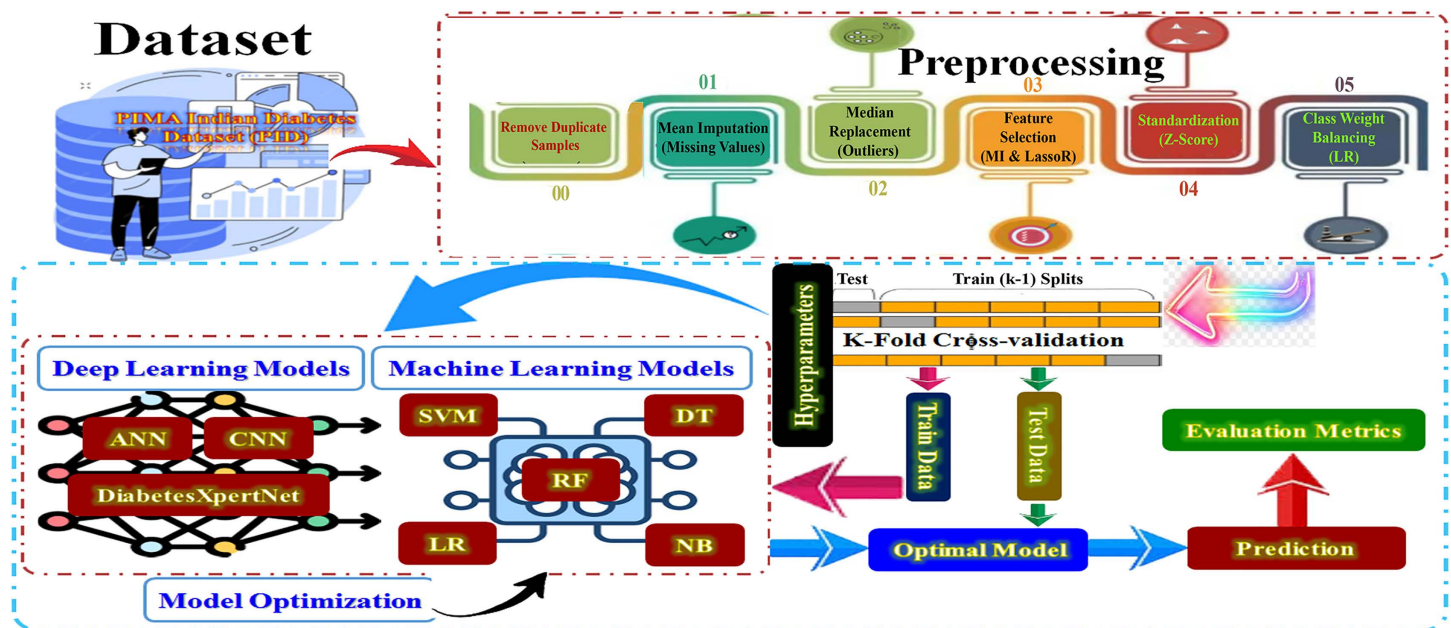


Fig 3. Robust framework for automatic and accurate diabetes prediction.

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integration of dynamic channel attention, C-AFE, and rigorous regularization methods. This carefully designed pipeline enhances predictive accuracy and reliability, providing a robust tool for medical data analysis. Comprehensive hyperparameter tuning and cross-validation further optimize model performance, contributing to improved diagnostics and outcomes in diabetes research.

3.2.1. Data preprocessing. Data preprocessing ensures high-quality input for T2D prediction, enhancing model performance (Fig 3). The pipeline includes the following steps:

(0) Remove Duplicate Samples

Duplicate samples are removed to prevent model bias and overfitting [47].

(1) Missing Value Detection and Imputation

Addressing missing and erroneous data is crucial for ensuring the accuracy and robustness of MLMs [48]. Missing values arise when certain features in a dataset lack complete information, potentially skewing analysis and impacting predictive performance [49]. Zero values in specific features are imputed using class-specific means. For a given feature value x_{ij} of the i -th sample and the j -th feature (where j corresponds to features such as blood pressure, insulin, BMI, skinfold thickness, and glucose), with k representing the class label (which is either 0 or 1 for the i -th sample), the imputation function $f(x_{ij}, k)$ is defined as follows (Eq. 1):

$$f(x_{ij}, k) = \begin{cases} \text{mean}(x_{ij}, k) & \text{if } x_{ij} = 0 \\ x_{ij} & \text{Otherwise} \end{cases} \quad (1)$$

where $\text{mean}(x_{ij}, k)$ is defined as follows (Eq. 2):

$$\text{mean}(x_{ij}, k) = (\sum_{t=1}^{N_k} x_{tj}) / N_k \quad (2)$$

Where x_{ij} represents the nonzero values of the j -th feature of the i -th sample with class label $k \in (0, 1)$ and where N_k indicates the number of nonzero values of the j -th feature of class k . This method ensures that class-specific data features are preserved while missing values are filled based on the average of observed values within the same class.

(2) Outlier detection and imputation

Outliers, data points significantly deviating from the expected range, may affect model performance and statistical analysis [29,50]. Their identification and treatment are crucial for reliable data processing. Outliers are identified using IQR bounds (Eq. 3).

$$L = Q_1 - (3/2) \times IQR, U = Q_3 + (3/2) \times IQR, IQR = Q_3 - Q_1 \quad (3)$$

L and U represent the lower and upper bounds for identifying outliers, with Q_1 and Q_3 being the first and third quartiles, respectively. The interquartile range (IQR), the difference between Q_3 and Q_1 , determines the spread of the middle 50% of the data and identifies the outlier threshold. Outliers in the data were identified and replaced with the median of the nonoutlier data points in the same feature. This approach ensures that the imputed values maintain the central tendency of the data while minimizing the impact of extreme values. The imputation function is defined as follows (Eq. 4):

$$g(x_{ij}) = \begin{cases} \text{median}(x_{ij}) & \text{if } x_{ij} \notin [L, U] \\ x_{ij} & \text{otherwise} \end{cases} \quad (4)$$

Where x_{ij} represents the value of the j -th feature in the i -th sample, the process of median imputation is defined as follows (Eq. 5):

$$\text{median}(x_{ij}) = \text{median}(x_{ij}) \quad \{\forall x_{ij} \in [L, U]\} \quad (5)$$

Replacing outliers with the median of valid data points minimizes distortion in the dataset, preserves the data distribution, and ensures robust model performance.

(3) Data standardization

In this study, features are standardized via Z-score (Eq. 6) [51]:

$$Z(x_{ij}) = (x_{ij} - M_j) / D_j \quad (6)$$

where $Z(x_{ij})$ represents the standardized value for the feature j -th and i -th observations. x_{ij} is the original value of the j -th feature for the i -th observation. Moreover, M_j is the mean of the j -th feature across the dataset, and D_j represents the standard deviation of the j -th feature.

(4) Feature Importance and Selection

Feature selection is essential in preprocessing to minimize dimensionality and optimize model performance [52]. A hybrid method integrating mutual information (MI) and LASSO regression with 5-fold cross-validation is employed to pinpoint critical features.

MI, a robust statistical metric, measures the dependency between features and the target variable using the SelectK-Best algorithm in Python [53]. Features are ranked by their relevance, with results compiled into a structured data frame for clarity.

Selection is further refined using LASSO regression, which applies an L_1 penalty to eliminate less significant coefficients, retaining only the most impactful features to mitigate overfitting [54,55]. The regularization parameter α is fine-tuned through cross-validation for optimal feature selection.

To enhance model robustness, correlations among selected features are evaluated via a correlation matrix, visualized with a spectral color map to highlight relationships and address multicollinearity. This integrated MI and LASSO approach ensures the identification of key features while simplifying the model.

(5) Weighted class balancing

Class imbalance can bias model predictions toward the majority class, underrepresenting the minority class. As shown in Figs 3 and 4, both datasets exhibit significant imbalance: Fig 4 shows the PID dataset with 500 non-diabetic patients (class 0) versus 268 diabetic patients (class 1), while Fig 5 displays the FHGD dataset with 1316 non-diabetic versus 684 diabetic patients. This imbalance poses challenges for achieving accurate and fair model outcomes, particularly in minority class detection.

To mitigate this, a class weighting approach using LR is employed. Minority class weights, ranging from 0 to 1 in 0.0005 increments, are optimized via GridSearchCV with 10-fold StratifiedKFold cross-validation to ensure robust tuning.

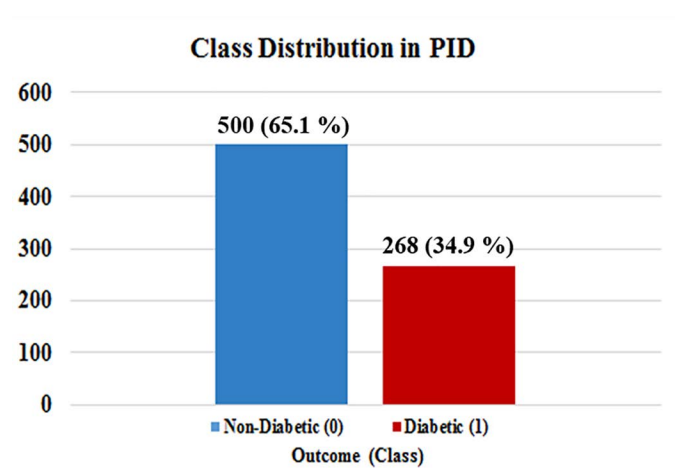


Fig 4. Bar chart showing the distribution of classes within the PID dataset. The chart highlights the imbalance in class representation, necessitating appropriate methods to address potential biases in model training, testing, and evaluation.

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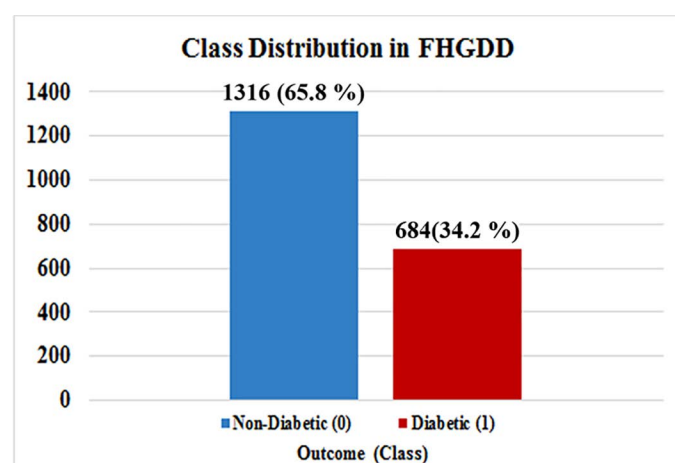


Fig 5. Bar chart showing the distribution of classes within the FHGD dataset. The chart highlights the imbalance in class representation, necessitating appropriate methods to address potential biases in model training, testing, and evaluation.

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3.2.2. k-fold cross-validation. A popular statistical technique for evaluating and contrasting MLM performance is cross-validation (CV) [56]. Separate data subsets are employed: one for training the model and the other for testing or validation. The dataset is partitioned into k equal (or equal) folds for k -fold cross-validation (kCV). The algorithm is then trained and validated k times, with each fold serving as the validation set while the remaining $k-1$ folds are used for training [56].

In the approach, a two-level cross-validation strategy was implemented. The inner loop involved 3-fold cross-validation, dedicated to hyperparameter tuning, employing the grid search method. After identifying the optimal hyperparameters, the model's effectiveness was assessed in the outer loop using 10-fold cross-validation. The procedure was repeated multiple times, as shown in Fig 6, to ensure the stability and reliability of the results.

3.2.3. Machine learning models. For binary classification, a variety of MLMs were used in this work, including Gaussian Naive Bayes (GNB) [51], LR [33], SVM [57], DT, and RF [51]. These models were chosen because they can reliably and accurately categorize data into two classes.

Models hyperparameters are optimized using grid search to explore various combinations and identify optimal configurations. Definitions and descriptions of the hyperparameters are provided in Table 3.

3.2.4. Deep learning models. This section outlines the DLMs used for T2DM diagnosis, evaluated on PID and FHGDD datasets for robustness. We first present the Multilayer Perceptron (MLP), adept at modeling complex relationships. Next, a tailored CNN is described, designed to extract hierarchical and spatial features from medical data. Finally, DiabetesXpertNet, a novel CNN-based model, integrates DCAM and C-AFE to prioritize clinically relevant features, enhancing accuracy and interpretability. Subsequent subsections detail each model's architecture and contributions to T2DM prediction, emphasizing DiabetesXpertNet's innovative design.

(1) Multilayer Perceptron:

MLPs are core DLMs with interconnected neurons linked by weights, serving as trainable parameters [58]. The MLP architecture used in this study features an input layer, multiple hidden layers, and an output layer [51]. The model processes an input vector with dimensionality D through these layers to produce an output vector with dimensionality N . For a neuron i in layer k , the output z_i^k is defined as follows (Eq. 7):

$$z_i^{(k)} = \sigma \left(\sum_j w_{ij}^{(k)} x_j^{(k-1)} + b_i^{(k)} \right) \quad (7)$$

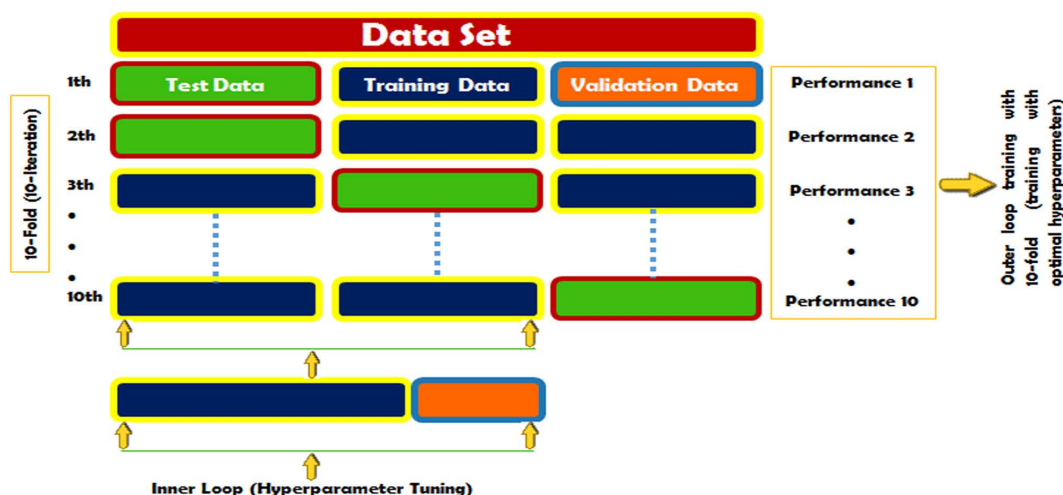


Fig 6. Partitioning of the PID and FHGD datasets for k -fold cross-validation in both hyperparameter tuning and evaluation.

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Table 3. Hyperparameter Specifications for MLMs Optimized via Grid Search in the Inner Loop.

MLMs	Hyperparameters	Description
LR	C	The regularization parameter. A smaller value specifies stronger regularization, whereas a larger value specifies weaker regularization.
	Penalty	The type of regularization used. It can be L_1 (L_1 regularization), L_2 (L_2 regularization), elasticnet (Elastic Net regularization), or none (no regularization).
	Solver	The optimization algorithm used to find the optimal parameters. It can be newton-cg (Newton conjugate gradient), lbfgs (Limited-memory BFGS), liblinear (Library for large-scale linear classification), or saga (Stochastic Average Gradient descent).
SVM	C	The regularization parameter. A smaller value specifies a larger margin but a higher error, whereas a larger value specifies a smaller margin but lower error.
	Kernel	The type of kernel function used to transform the data. It can be linear, poly, rbf, or sigmoid kernels.
	Gamma	The kernel coefficient for rbf, Polynomial, and Sigmoid kernels.
DT	Splitter	The strategy used to choose the split at each node. It can be best (find the best split) or random (choose a random split).
	Min_samples_split	The least amount of samples needed to divide an internal node.
	Min_samples_leaf	The minimum number of samples required for a node to be a leaf.
	Criterion	The function that assesses a split's quality. It could be either entropy (information gain) or gini (Gini impurity).
RF	N_estimators	The quantity of trees within the woodland. Whereas raising this value generally enhances the model's performance, it also raises the computational expense.
	Max_depth	each tree's maximum depth. It can be either None (nodes are enlarged until they contain fewer samples than min_samples_split samples) or an integer value indicating the forest's maximum tree depth.
	Min_samples_split	The least amount of samples needed in the forest's trees in order to separate an internal node.
	Min_samples_leaf	The bare minimum of samples needed for a node to resemble a leaf on a forest tree.
	Max_features	the quantity of features to take into account in each tree whereas determining the ideal split. The values for auto, sqrt (n_features), and log2 (base 2 logarithm of the number of features) can be substituted for sqrt (n_features).
	Criterion	The measuring function used to evaluate the quality of a split in the trees. It can be gini (Gini impurity) or entropy (information gain).
GNB	Var_smoothing	a proportion of the maximum variance among all features.

<https://doi.org/10.1371/journal.pone.0330454.t003>

Where $x_j^{(k-1)}$ represents the input values from the previous layer and $w_{ij}^{(k)}$ represents the weights. The bias is denoted by $b_i^{(k)}$ and represents the bias, and the activation function is represented by σ , which may be a ReLU, sigmoid, or tanh function.

The training process employs the backpropagation algorithm to refine the weights. The error δ is defined as follows (Eq. 8):

$$\delta = y_{true} - y_{pred} \quad (8)$$

Where y_{true} is the actual value and y_{pred} is the predicted value generated by the model. Weight adjustments are made according to the following update rule (Eq. 9):

$$w_{ij}^{(k)} = w_{ij}^{(k)} + \eta \cdot \delta \cdot x_j^{(k-1)} \quad (9)$$

Where η represents the learning rate, which controls the step size of weight updates during training.

Determining the optimal number of hidden layers and neurons in each layer is crucial, as these hyperparameters influence model performance. Adding more layers and neurons allows the model to learn more intricate patterns from the data, but it also increases the need for a larger dataset to avoid overfitting.

Fig 7 illustrates the architecture of the MLP utilized in this study, featuring (D)-dimensional input layers, (m) hidden layers, and an output layer with two neurons for classifying nondiabetic (0) and diabetic (1) patients. The input data are characterized by a dimensionality of (D), where (D) is derived from the feature selection method outlined in this study. The network architecture comprises (m) hidden layers, with m_i denoting the number of neurons in the (i)-th hidden layer. The output layer comprises two neurons corresponding to the classification labels: nondiabetic (0) and diabetic (1) patients.

Hyperparameters are optimized via grid search, with definitions provided in Table 4. The MLP is evaluated on PID and FHGD datasets for robust performance.

(2) Convolutional neural network model architecture:

CNNs are advanced feed-forward neural networks designed for processing grid-structured data. Unlike traditional neural networks, CNNs excel at feature extraction through their convolutional and pooling layers and perform classification via fully connected layers [31,59]. In this study, a CNN model tailored for predicting T2DM is introduced and evaluated on both the PID and FHGD datasets (Fig 8).

The process begins with an input layer that accepts data formatted with a specific feature shape. The model starts with a 1D convolutional layer using the ReLU activation function and 32 filters, each with a kernel size of 3. This setup enables the model to extract basic features from the input data. The spatial dimensions of the feature maps are then reduced by an average pooling layer with a

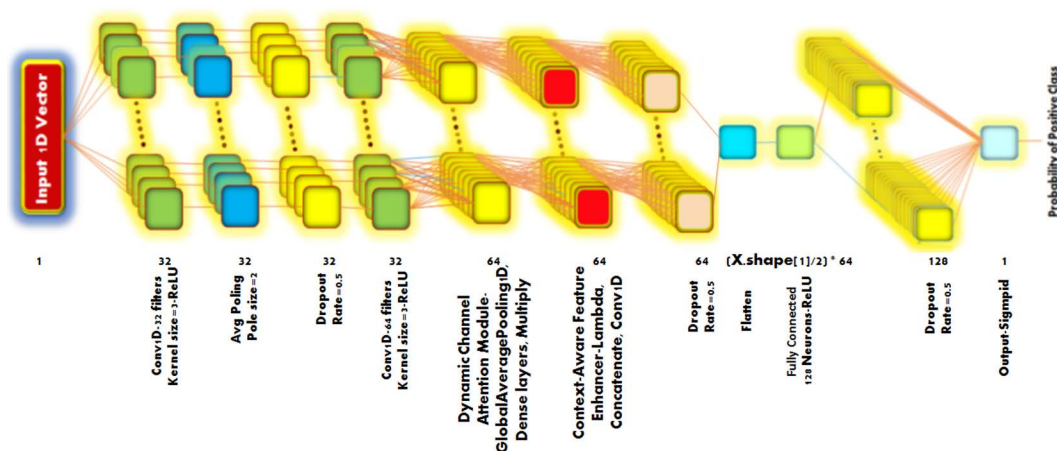


Fig 7. Architecture of the ANN used in this study, featuring D -dimensional input layers, m hidden layers, and an output layer with two neurons for classifying nondiabetic (0) and diabetic (1) patients.

<https://doi.org/10.1371/journal.pone.0330454.g007>

Table 4. Hyperparameter Specifications for MLP DLM Optimized via Grid Search in the Inner Loop.

DLM	Hyperparameters	Description
MLP	Number of Hidden Layers	The quantity of network's hidden layers.
	Neurons per Layer	The quantity of neurons found in every hidden layer.
	Activation Function	The activation function used in each hidden layer (e.g., ReLU, Sigmoid, Tanh).
	Neuron_INITIALIZER	The method for initializing the weights of neurons (e.g., He normal, Glorot uniform).
	Learning Rate (η)	The rate at which weights are updated during training.
	Batch Size	The quantity of samples used in each update of the weights.
	Number of Epochs	How many times the model is trained using the complete dataset.
	Dropout Rate	The proportion of neurons that are turned off at random whereas being trained.
	Loss Function	The function (such as MSE or Binary Cross-Entropy) that the model seeks to minimize.
	Optimizer	The algorithm used to optimize weights (e.g., SGD, Adam, RMSprop).

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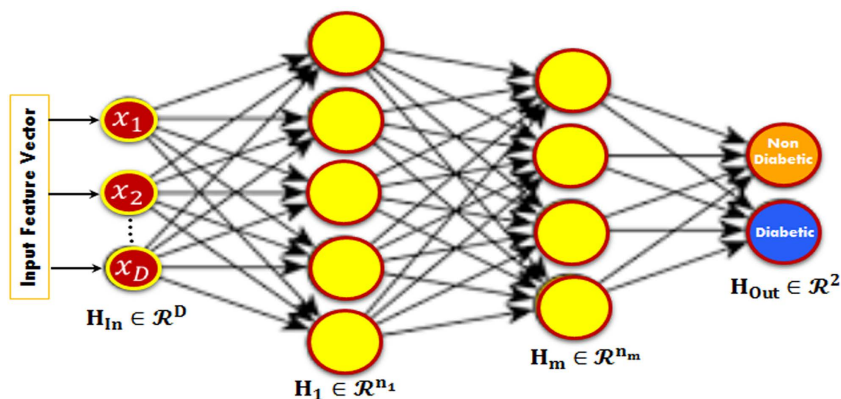


Fig 8. Detailed architectural diagram of the CNN model for predicting T2DM in the PID and FHGD datasets.

<https://doi.org/10.1371/journal.pone.0330454.g008>

pool size of 2, which helps maintain significant features while reducing computational complexity. A dropout layer with a 0.5 dropout rate is added to mitigate the danger of overfitting. By randomly deactivating certain neurons during training, dropout serves as a regularization strategy that improves the model's ability to generalize to new, unknown data. A second 1D convolutional layer with a ReLU activation function and 64 filters, using the same kernel size, completes the model. This deeper convolutional layer extracts more intricate information from the data. To further reduce overfitting, a second dropout layer with a rate of 0.5 is added [60].

After flattening the feature maps into a one-dimensional vector, a fully connected layer with 128 neurons and ReLU activation processes the vector. The retrieved features are combined into a comprehensive representation in this layer. A final dropout layer with a rate of 0.5 is added to the output of the fully connected layer to further prevent overfitting. The output layer of the model, which has a single neuron with a sigmoid activation function, generates a probability score indicating the likelihood of the input data belonging to the positive class (i.e., diabetes). The structure and operation of the CNN model are summarized in Table 5. This design leverages feature extraction, dimensionality reduction, and classification to deliver precise predictions for diabetes diagnosis within the dataset.

(3) DiabetesXpertNet: An innovative Approach for T2DM Prediction:

DiabetesXpertNet represents a pioneering advancement in convolutional neural network-based models for T2D prediction, achieving exceptional accuracy and robustness on the PID dataset and FHGD datasets. Unlike CNNs and attention-based models, such as SE-Net [22] and CBAM [23], which are primarily designed for image data and rely on generalized channel and spatial attention mechanisms, DiabetesXpertNet is specifically tailored for tabular medical data. By directly processing structured datasets without requiring transformation into image formats (e.g., as in [29]), it preserves feature granularity and reduces preprocessing overhead. The model incorporates a DCAM and a C-AFE to prioritize clinically relevant features, such as glucose and insulin levels, and enhance interpretability for medical applications.

Fig 9 illustrates the architecture of DiabetesXpertNet, and Table 6 details its layer configuration.

Fig 9 provides a detailed architectural diagram of the DiabetesXpertNet model, designed specifically for T2D prediction using the PID dataset and FHGD datasets. Table 6 outlines the key components of this architecture. The model leverages a GlobalAveragePooling1D layer to aggregate comprehensive global information, followed by dense layers to assess channel significance. A recalibration mechanism dynamically adjusts feature weights, emphasizing critical attributes while minimizing irrelevant noise, ensuring enhanced predictive accuracy and consistency.

DCAM: This innovative module refines feature representation by dynamically prioritizing the most relevant channels during training and inference. Unlike SE-Nets [22], which focus on channel-wise recalibration for image data, and CBAM [23], which combines channel and spatial attention for image processing, this module is optimized for tabular data. It employs GlobalAveragePooling1D to aggregate global feature information, followed by dense layers to evaluate channel significance, and a multiplication operation

Table 5. Detailed Layer Configuration of the CNN Model for T2DM Prediction.

Type of Layer	Setup	Shape of the Input	Shape of the Output	Goal
Input Layer	–	(None, X.shape[1], 1)	(None, X.shape[1], 1)	Accepts the input data and defines the input shape for the model.
1D Convolutional Layer	32 filters, 3 kernel sizes, “relu” activation, and “same” padding.	(None, X.shape[1], 1)	(None, X.shape[1], 32)	Extracts local features from the input data, with 32 filters capturing different patterns.
Average Pooling Layer	Pool size=2.	(None, X.shape[1], 32)	(None, X.shape[1]/2, 32)	Reduces the spatial dimension, retaining key features whereas decreasing computational complexity.
Dropout Layer	Dropout rate=0.5.	(None, X.shape[1]/2, 32)	(None, X.shape[1]/2, 32)	Prevents overfitting by randomly dropping 50% of the neurons during training.
1D Convolutional Layer	64 filters, 3 kernel sizes, “relu” activation, and “same” padding.	(None, X.shape[1]/2, 32)	(None, X.shape[1]/2, 64)	Further extracts features with increased filter depth, capturing more complex patterns.
Dropout Layer	Dropout rate=0.5.	(None, X.shape[1]/2, 64)	(None, X.shape[1]/2, 64)	Mitigates overfitting by applying dropout to the layer.
Flatten Layer	–	(None, X.shape[1]/2, 64)	(None, (X.shape[1]/2) * 64)	Transforms the fully connected layer's 2D feature maps into a 1D vector.
Fully Connected Layer	128 units, “relu” activation.	(None, (X.shape[1]/2) * 64)	(None, 128)	Learns complex representations by combining features into 128 new dimensions.
Dropout Layer	Dropout rate=0.5.	(None, 128)	(None, 128)	Reduces overfitting in the fully connected layer by randomly deleting neurons during training.
Output Layer	1 unit, “sigmoid” activation	(None, 128)	(None, 1)	Provides the likelihood that an input will fall into the positive class and be appropriate for binary classification.

<https://doi.org/10.1371/journal.pone.0330454.t005>

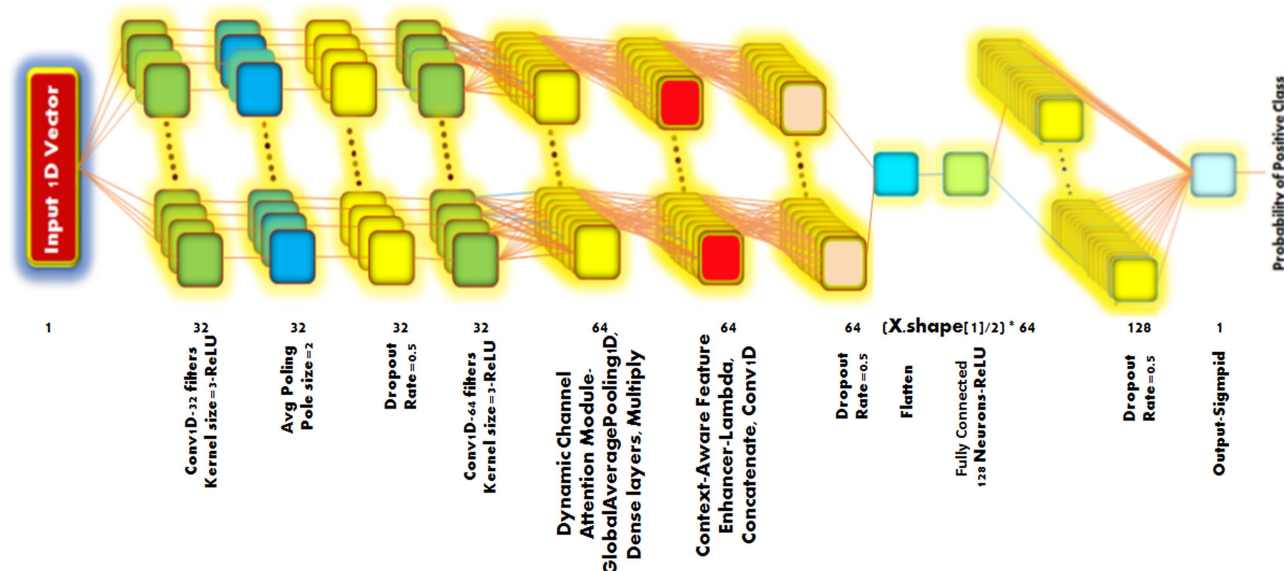


Fig 9. Detailed architectural diagram of the DiabetesXpertNet model for predicting T2DM in the PID and FHGD datasets.

<https://doi.org/10.1371/journal.pone.0330454.g009>

to recalibrate feature channels based on their importance (e.g., prioritizing clinically relevant features like glucose and insulin). This targeted mechanism enhances the model's focus on critical features, reducing the influence of less significant attributes and improving prediction accuracy compared to conventional CNNs, which lack such tailored feature refinement.

Table 6. Detailed Layer Configuration of the DiabetesXpertNet Model for T2DM Prediction.

Type of Layer	Setup	Shape of the Input	Shape of the Output	Goal
Input Layer	–	(None, X.shape [1], 1)	(None, X.shape [1], 1)	Accepts the input data and define the input shape for the model.
1D Convolutional Layer	32 filters, 3 kernel sizes, “relu” activation, and “same” padding.	(None, X.shape [1], 1)	(None, X.shape [1], 32)	Extracts local features from the input data, with 32 filters capturing distinct patterns.
Average Pooling Layer	Pool size = 2.	(None, X.shape [1], 32)	(None, X.shape [1]/2, 32)	Reduces the spatial dimension, keeping key features, whereas decreasing computational complexity.
Dropout Layer	Dropout rate = 0.5.	(None, X.shape[1]/2, 32)	(None, X.shape[1]/2, 32)	Prevents overfitting by dropping 50% of the neurons during training.
1D Convolutional Layer	64 filters, 3 kernel sizes, “relu” activation, and “same” padding.	(None, X.shape [1]/2, 32)	(None, X.shape[1]/2, 64)	Further extracts features with increased filter depth, capturing more complex patterns.
Dropout Layer	Dropout rate = 0.5.	(None, X.shape[1]/2, 64)	(None, X.shape[1]/2, 64)	Mitigates overfitting by applying dropout to the layer.
Dynamic Channel Attention Module	GlobalAveragePooling1D, Dense layers, Multiply.	(None, X.shape[1]/2, 64)	(None, X.shape[1]/2, 64)	Enhances feature representation by focusing on the most relevant channels via attention mechanism.
Context-Aware Feature Enhancer	Lambda, Concatenate, Conv1D.	(None, X.shape[1]/2, 64)	(None, X.shape[1]/2, 64)	Amplifies important features by applying spatial attention considering the context of features.
Flatten Layer	–	(None, X.shape[1]/2, 64)	(None, (X.shape[1]/2) * 64)	Transforms the fully connected layer’s 2D feature maps into a 1D vector.
Fully Connected Layer	128 units, “relu” activation.	(None, (X.shape[1]/2) * 64)	(None, 128)	Learns complex representations by combining features into 128 additional dimensions.
Dropout Layer	Dropout rate = 0.5.	(None, 128)	(None, 128)	Reduces overfitting in the fully connected layer by deleting neurons during training.
Output Layer	1 unit, “sigmoid” activation	(None, 128)	(None, 1)	Provides the likelihood that an input will fall into the positive class and be appropriate for binary classification.

<https://doi.org/10.1371/journal.pone.0330454.t006>

C-AFE: The C-AFE introduces an advanced mechanism to amplify key features by capturing their contextual relationships within tabular data. Unlike the spatial attention mechanisms in SE-Net s [22] and CBAM [23], which are designed for image data, this module uses one-dimensional convolutional layers, Lambda, and Concatenate operations to model sequential dependencies in structured datasets. For example, it captures relationships between clinical features across patient records, enhancing predictive performance. By avoiding the need for data transformation into images, as required in some convolutional approaches [28], this module preserves feature integrity and reduces computational overhead, resulting in superior performance compared to traditional CNNs that struggle with non-spatial data structures.

In summary, the integration of a DCAM and a C-AFE establishes DiabetesXpertNet as a robust and precise tool for T2DM. With approximately 1.2 million parameters, it is more efficient than attention-based models like CBAM (~1.5 million parameters), offering enhanced interpretability critical for clinical decision-making. These innovations distinguish DiabetesXpertNet as a groundbreaking solution for addressing the complexities of medical data analysis.

4. Evaluation metrics

Model performance for distinguishing nondiabetic (class 0) and diabetic (class 1) patients is evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC, defined as follows:

(1) Accuracy: Proportion of correct classifications. It is defined as follows (Eq. 10):

$$\text{Accuracy} = (TP + FN) / (TP + TN + FP + FN) \quad (10)$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives

(2) Precision: Percentage of correctly identified diabetic patients among predicted positives. Its definition is defined as follows (Eq. 11):

$$\text{Precision} = TP / (TP + FP) \quad (11)$$

(3) Recall: Proportion of actual diabetic patients correctly identified. It is defined as follows:

$$\text{Recall} = TP / (TP + FN) \quad (12)$$

(4) F1 score: Harmonic mean of precision and recall. It is defined as follows (Eq. 13):

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (13)$$

(5) AUC-ROC: Area under the receiver operating characteristic curve, plotting True Positive Rate (Recall) against False Positive Rate. The AUC is defined as follows (Eq. 14):

$$\text{AUC} = \int_0^1 \text{Recall}(\text{FPR}) d\text{FPR} \quad (14)$$

In this formula, $d\text{FPR}$ represents the infinitesimal change in FPR along the x-axis of the ROC curve. The integral accumulates these small areas under the curve to compute the total area, providing a comprehensive measure of the model's performance across all possible threshold values. TPR is equivalent to Recall and FPR is the ratio of incorrectly predicted positive instances to the total number of actual negative instances and is defined as follows (Eq. 15):

$$\text{FPR} = FP / (FP + TN) \quad (15)$$

AUC measures discriminative ability, with values near 1 indicating strong performance.

5. Computational setup

Experiments were conducted on Google Colab Premium with a TPU, 96 virtual cores, and 334 GB RAM on a Linux system (kernel: #1 SMP PREEMPT_DYNAMIC, June 27, 2024). Software includes Python 3.10.12, NumPy 1.26.4, TensorFlow 2.12, Keras 2.12, and Scikit-learn 1.3.2 for numerical, deep learning, and ML tasks.

6. Results

This section describes the experiments conducted to evaluate the proposed methods. The impact of missing data and outlier imputation is analyzed, assessing how imputation strategies influence model performance. Feature importance and selection processes are examined for their effect on predictive accuracy. The role of weighted class balancing in enhancing robustness against class imbalances is evaluated. Performance of MLMs and DMMs is compared across diverse datasets. These experiments outline the methodology to assess the proposed techniques.

6.1. Data preprocessing

This section outlines the methods used to preprocess datasets. The procedures include removing duplicate samples, imputing missing values and outliers, selecting important features, and applying weighted class balancing to enhance model fairness and robustness.

(0) Remove Duplicate Samples:

As an initial step, duplicate samples are removed from the datasets. The PID dataset has no duplicates, retaining all 768 samples. The FHGD dataset, with 2000 samples, contains 1256 duplicates, reduced to 744 unique samples after removal. This ensures distinct data points for analysis, minimizing redundancy.

(1) Impact of Missing Data and Outlier Imputation

This study employs detection and imputation methods for missing values in the PID and FHGD datasets, as described in the preprocessing section 3.2.1. Zero values are identified in glucose, blood pressure, skin thickness, insulin, and BMI features (Table 7), with skin thickness and insulin showing the highest frequency of missing entries. The pregnancies feature, reflecting the number of pregnancies, includes zero entries for individuals with no pregnancies, which are not imputed to preserve data integrity.

Outliers in the PID and FHGD datasets are identified using the method described in the preprocessing section 3.2.1 for features including pregnancies, insulin, BMI, skin thickness, age, blood pressure, and diabetes pedigree function. Kernel density estimation (KDE) and box plots, shown in Fig 10 for PID and Fig 12 for FHGD datasets, display these outliers. Instead of removal, outliers are replaced with median values of the respective features to preserve dataset integrity and sample size. Post-imputation, Fig 11 for PID and Fig 13 for FHGD illustrate the adjusted attribute distributions.

Figs 14, 15–17 provide insights through correlation heatmaps for the PID and FHGD datasets. For PID dataset, Fig 14 shows the correlation matrix before missing value imputation, outlier replacement, and standardization, highlighting distortions due to incomplete data, extreme values, and unscaled features, while Fig 15 displays the matrix after these steps, revealing enhanced variable relationships. Similarly, for FHGD dataset, Fig 17 presents the correlation matrix before preprocessing, and Fig 16 shows improved relationships post-processing. For PID dataset, Pearson correlation coefficients changed significantly post-preprocessing: skin thickness with outcome increased from 0.07 to 0.31, blood pressure with outcome decreased from 0.70 to 0.18, and insulin with outcome rose from 0.13 to 0.52. For FHGD dataset, correlations shifted as follows: skin thickness with outcome increased from 0.08 to 0.20, blood pressure with outcome rose from 0.08 to 0.31, and insulin with outcome increased from 0.12 to 0.52. These results demonstrate that missing value imputation, outlier replacement, and standardization significantly improve data quality for both datasets.

The mean imputation strategy, described in section 3.2.1, is computationally lightweight and well-suited for real-time clinical environments where features like insulin may be unavailable due to cost or equipment limitations. For instance, insulin exhibited significant missing values (374 in PID, 359 in FHGD dataset, Table 7), which were replaced using class-specific means to preserve data integrity. This approach ensures reliable imputation during inference, enabling DiabetesXpertNet to maintain robust predictions in resource-constrained settings. Furthermore, the DCAM adaptively prioritizes available features, such as glucose and BMI, which have strong predictive power (e.g., glucose-insulin correlation of 0.52 for PID, Fig 15), enhancing the model’s ability to handle missing data effectively in clinical practice.

Table 7. Attribute Wise Count of Erroneous or Missing Values in the PID and FHGD Datasets.

Attributes	Count of Missing Values	
	PID dataset	FHGD dataset
Pregnancies	111	112
Glucose	5	5
Blood Pressure	35	34
Skin Thickness	227	215
Insulin	374	359
BMI	11	10
Diabetes Pedigree Function	0	0
Age	0	0

<https://doi.org/10.1371/journal.pone.0330454.t007>

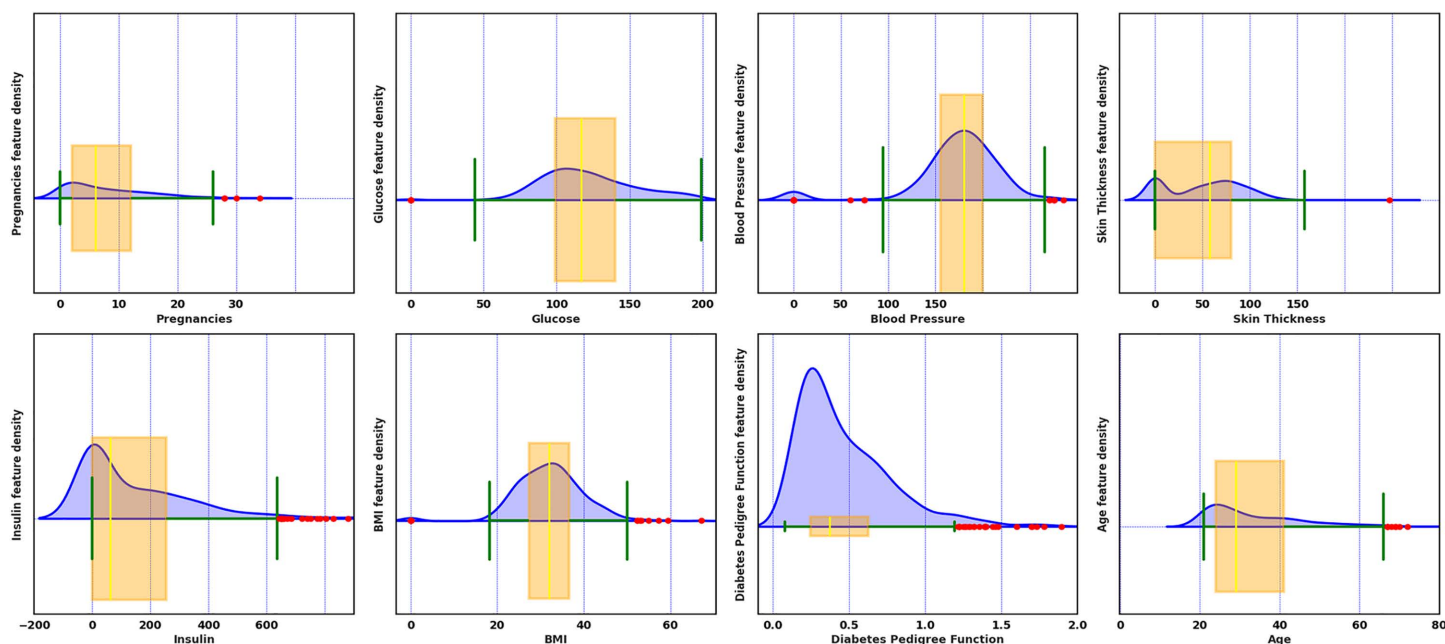


Fig 10. Boxplot of PID dataset before outlier imputation. The plot shows the initial distribution of attributes prior to data cleaning.

<https://doi.org/10.1371/journal.pone.0330454.g010>

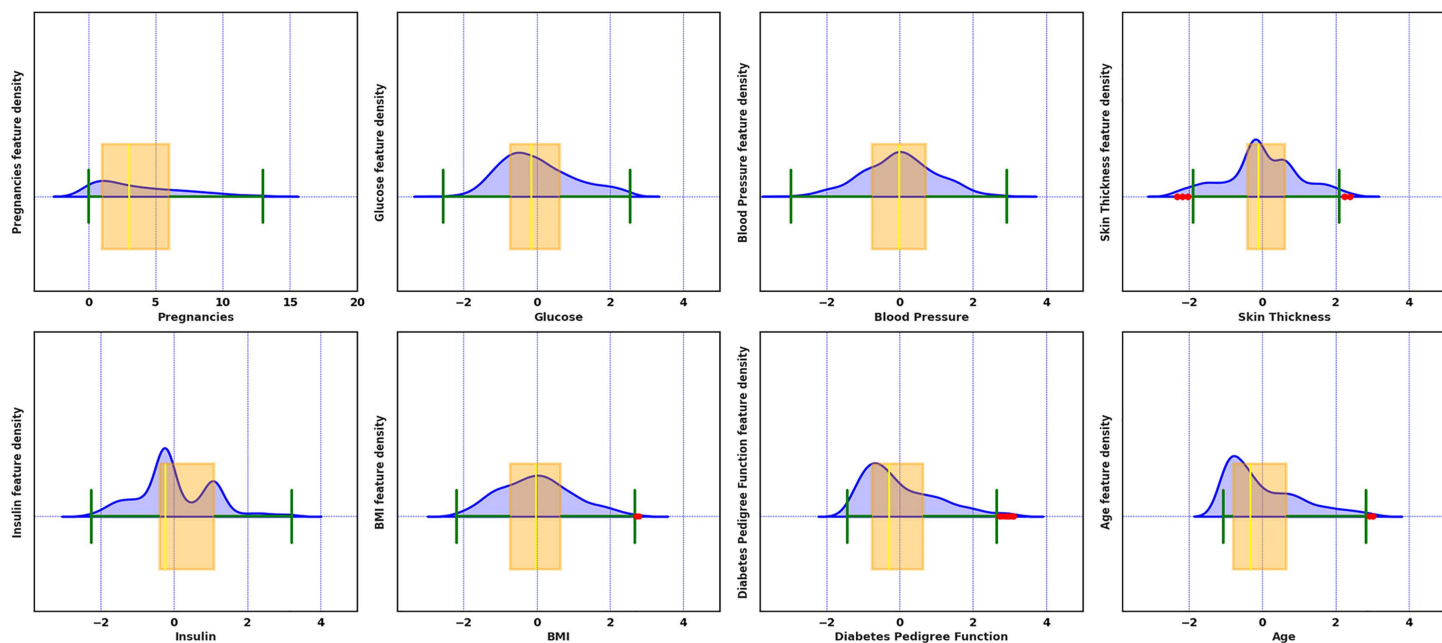


Fig 11. Boxplot of PID dataset after outlier imputation. The plot demonstrates the refined attribute distribution following outlier treatment.

<https://doi.org/10.1371/journal.pone.0330454.g011>

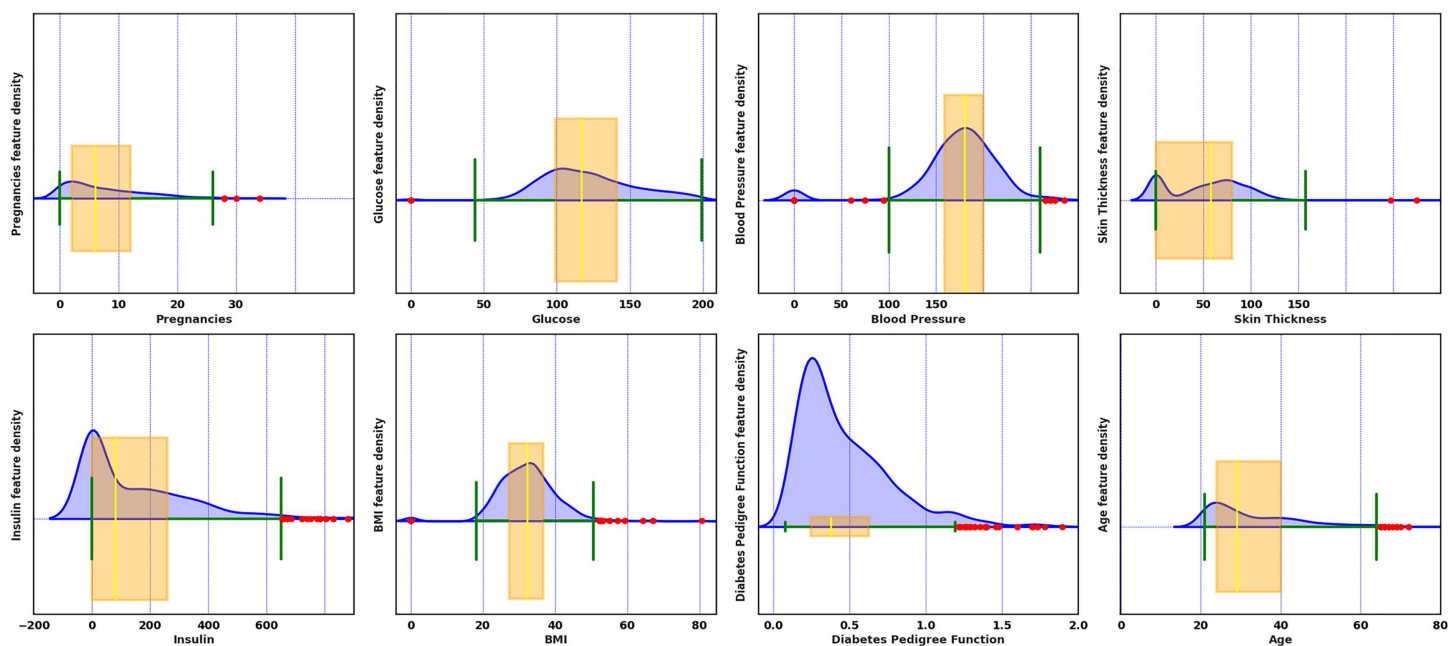


Fig 12. Boxplot of FHGD dataset before outlier imputation, showing initial distribution patterns and outlier presence across attributes.

<https://doi.org/10.1371/journal.pone.0330454.g012>

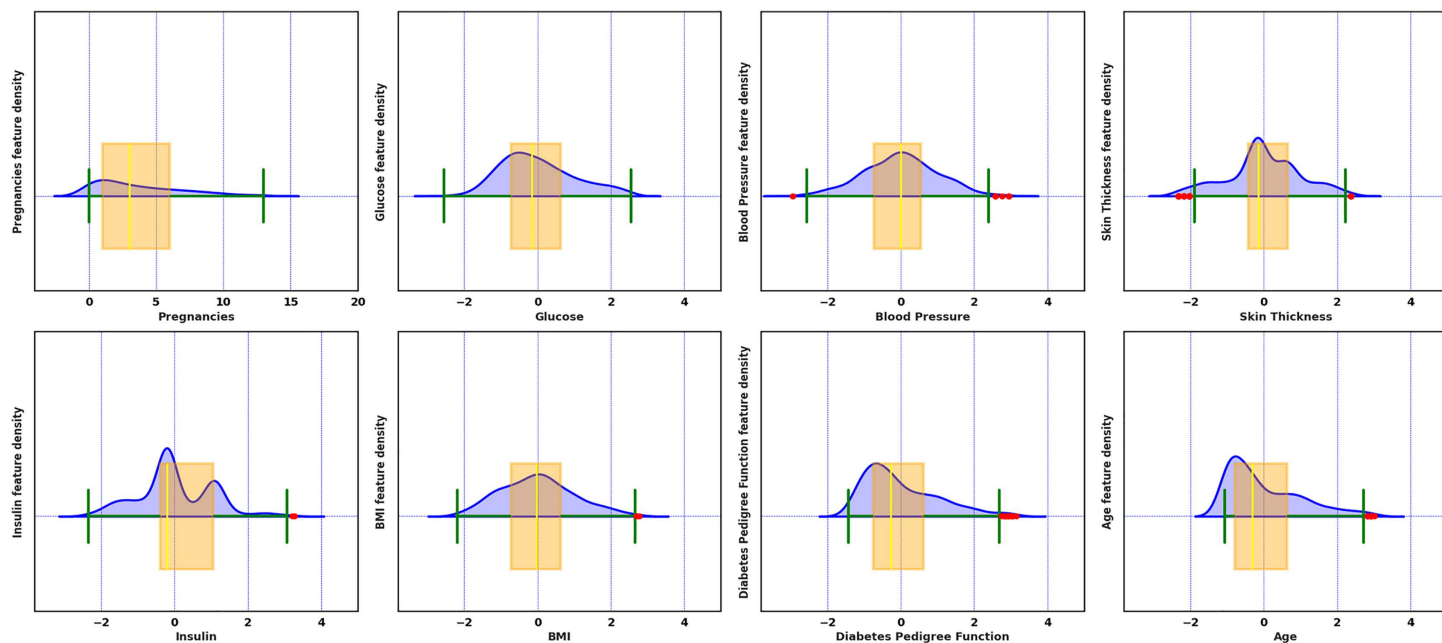


Fig 13. Boxplot of FHGD dataset after outlier imputation, demonstrating normalized distributions and the effectiveness of our data cleaning methodology.

<https://doi.org/10.1371/journal.pone.0330454.g013>

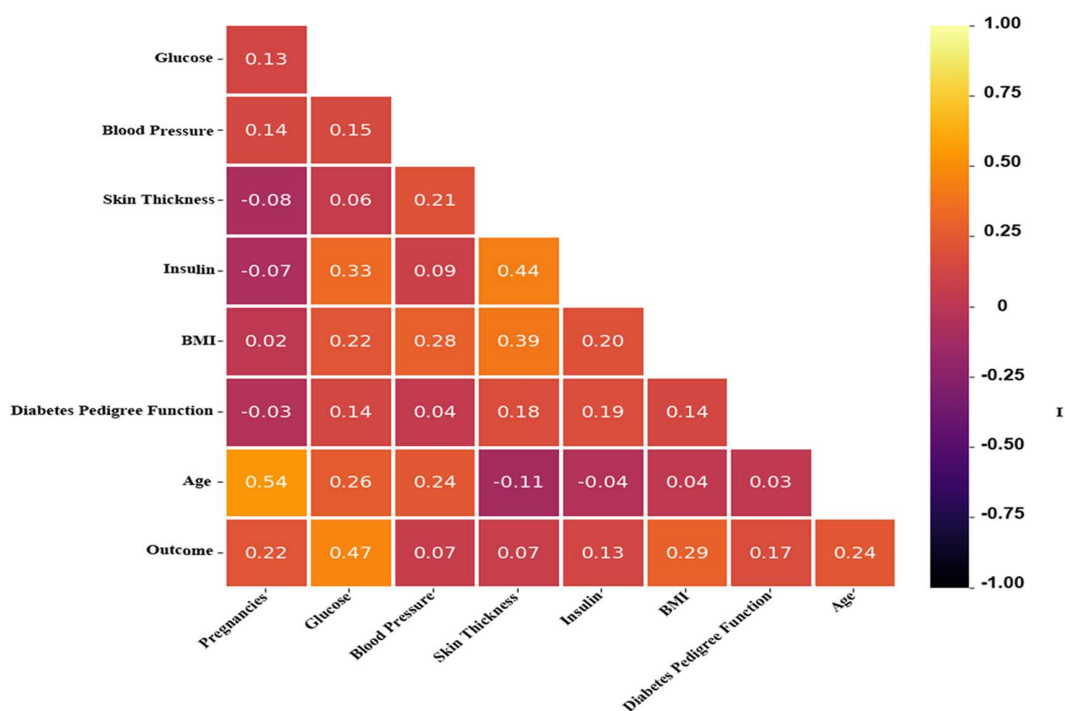


Fig 14. Confusion matrix for PID dataset before preprocessing, showing baseline classification performance of attributes against outcomes.

<https://doi.org/10.1371/journal.pone.0330454.g014>

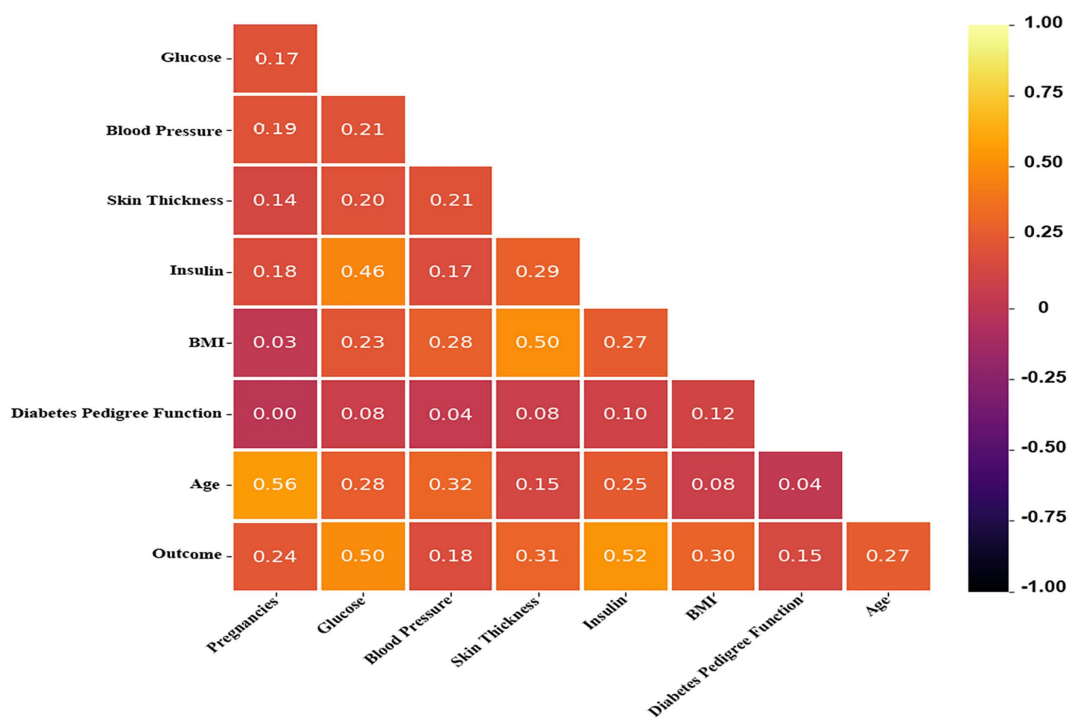


Fig 15. Confusion matrix for PID dataset after preprocessing, demonstrating improved classification accuracy resulting from data cleaning and transformation.

<https://doi.org/10.1371/journal.pone.0330454.g015>

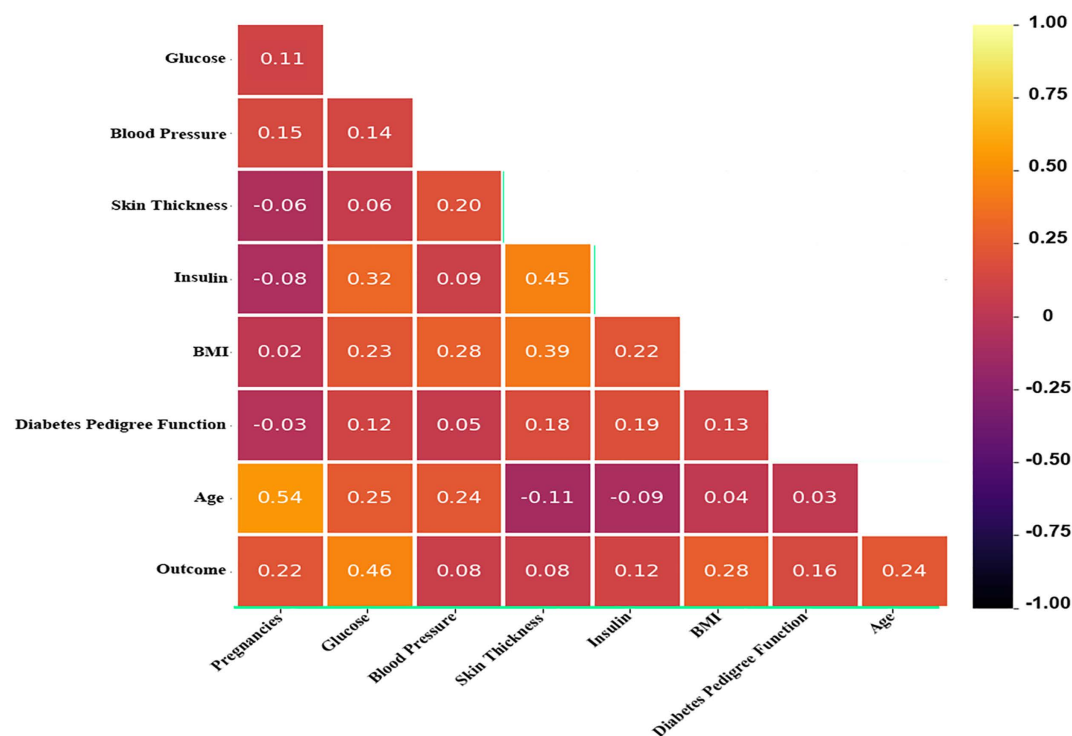


Fig 16. Confusion matrix for FHGD dataset before preprocessing.

<https://doi.org/10.1371/journal.pone.0330454.g016>

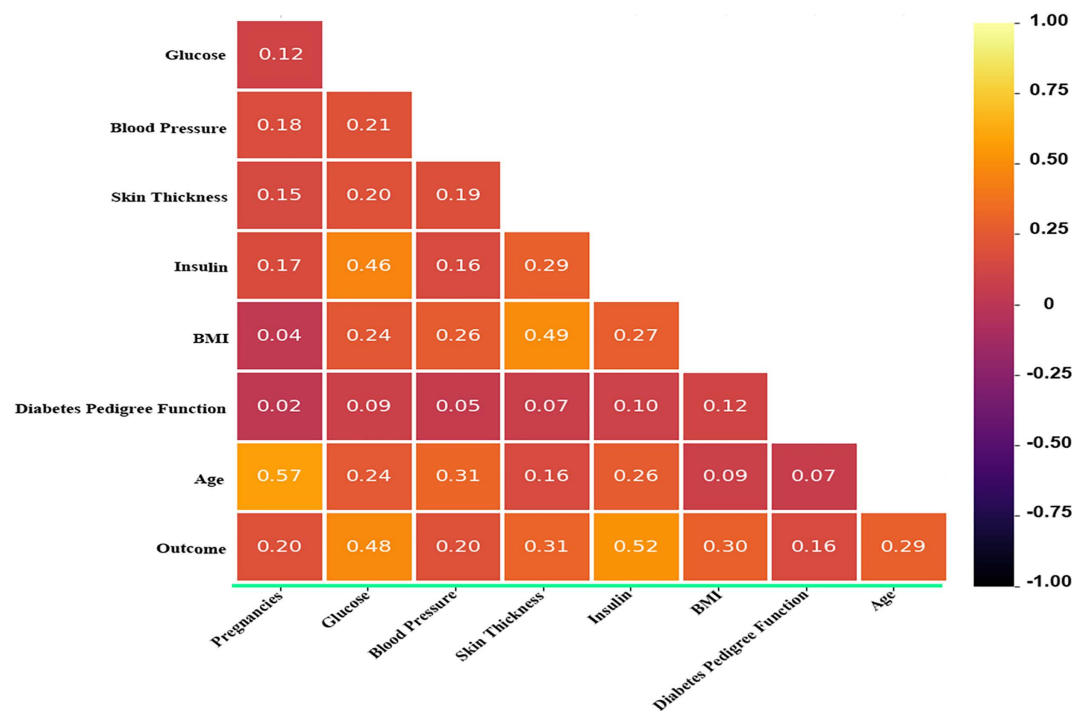


Fig 17. Confusion matrix for FHGD dataset after preprocessing.

<https://doi.org/10.1371/journal.pone.0330454.g017>

(2) Feature Importance and Selection Results

This section describes the feature selection process using a combined MI and LassoR approach, as outlined in the preprocessing section 3.2.1. Features are initially filtered based on MI scores, followed by refinement with LassoR. [Table 8](#) ranks features by MI scores, measuring their dependency on the target variable.

As shown in [Table 8](#), features for the PID dataset and FHGD dataset datasets were ordered according to their impact on diabetes risk based on MI scores. For PID, the top four features—glucose, skin thickness, insulin, and age—presented the highest MI scores and were selected for further analysis. For FHGD dataset, the top four features—glucose, skin thickness, insulin, and BMI—were similarly selected. After initial selection, LassoR refined these features for both datasets by applying an L1 penalty, shrinking less significant feature coefficients to zero, and selecting the most critical features. The final features selected by LassoR for each dataset are detailed in [Table 9](#).

As displayed in [Table 9](#), LassoR feature selection was applied to the PID dataset and FHGD dataset datasets to identify significant predictors of diabetes risk. For PID dataset, LassoR initially retained glucose, skin thickness, insulin, and age with nonzero coefficients, but with the regularization parameter α set to 0.1, only glucose and insulin were selected. Similarly, for FHGD dataset, LassoR retained glucose, skin thickness, insulin, and BMI initially, and with $\alpha=0.1$, only glucose and insulin were selected. These outcomes underscore the sensitivity of LassoR to the regularization parameter and highlight the importance of fine-tuning α to achieve optimal model configurations for both datasets.

As displayed in [Table 9](#), LassoR feature selection was applied to the PID and FHGD datasets to identify significant predictors of diabetes risk. For PID, LassoR retained glucose, skin thickness, insulin, and age with nonzero coefficients. For FHGD dataset, LassoR selected glucose, skin thickness, insulin, and BMI with nonzero coefficients. Adjusting the regularization parameter α to 0.1 reduced the selection to glucose and insulin for both datasets. These outcomes underscore the sensitivity of LassoR to the regularization parameter and highlight the importance of fine-tuning α to achieve optimal model configurations. All models considered in this research achieved their optimal performance when these four chosen features were used for each dataset, reinforcing the effectiveness of the feature selection process.

Table 8. Feature Importance Based on MI Scores.

Rank	PID dataset		FHGD dataset	
	Feature	MI Score	Feature	MI Score
1	Insulin	0.291068	Insulin	0.265522
2	Skin Thickness	0.183832	Skin Thickness	0.227881
3	Glucose	0.128101	Glucose	0.118784
4	Age	0.090331	BMI	0.069263
5	BMI	0.074983	Blood Pressure	0.045343
6	Pregnancies	0.058735	Age	0.043437
7	Blood Pressure	0.031197	Pregnancies	0.017667
8	Diabetes Pedigree Function	0.002046	Diabetes Pedigree Function	0.011735

<https://doi.org/10.1371/journal.pone.0330454.t008>

Table 9. Final Features Selected by LassoR.

Rank	PID dataset		FHGD dataset	
	Feature	LassoR Coefficient	Feature	LassoR Coefficient
1	Glucose	0.138985	Glucose	0.137570
2	Skin Thickness	0.067816	Skin Thickness	0.053609
3	Insulin	0.155276	Insulin	0.156497
4	Age	0.038344	BMI	0.037859

<https://doi.org/10.1371/journal.pone.0330454.t009>

The hybrid approach employed in this study, which combines MI and LassoR, proved effective in identifying the key features influencing diabetes risk. MI was used to select features most relevant to the target variable. LassoR was used to refine the selection by retaining only the most essential features. This procedure improved the accuracy of the predictive model, whereas also offering insightful information about the most important diabetes risk factors.

(3) Evaluation of the effect of weighted class balancing on model accuracy

To address class imbalance in the PID and FHGD datasets, a weighted class balancing technique, as described in section 3.2.1, was applied across all models evaluated in this study, including baseline MLMs (LR, SVM, DT, RF, GNB) and DLMs (DiabetesXpertNet, CNN, MLP). Using GridSearchCV with 10-fold cross-validation, class weights were optimized to maximize classification accuracy by assigning higher weights to the minority class (diabetic) to mitigate bias toward the majority class (non-diabetic). The fine-tuned weights, presented in Table 10, indicate an optimal distribution for PID dataset, with 0.4715 for the non-diabetic class (class 0) and 0.5285 for the diabetic class (class 1). Similarly, for FHGD dataset, the optimal distribution is 0.4735 for class 0 and 0.5265 for class 1. These balanced weightings were applied consistently across all models, ensuring uniform handling of class imbalance for both datasets

Figs 18 and 19, illustrates the precision-weight trade-off, with accuracy trends for non-diabetic patients (class 0) shown by the red dashed line, highlighting the critical role of precise weight calibration in enhancing model performance, as further detailed in Table 10 and Figs 18 and 19.

Table 10. Fine-Tuned Class Weights for Improved Classification Accuracy.

Class	Weight	
	PID dataset	FHGD dataset
Nondiabetic (class 0)	0.4715	0.4735
Diabetic (class 1)	0.5285	0.5265

<https://doi.org/10.1371/journal.pone.0330454.t010>

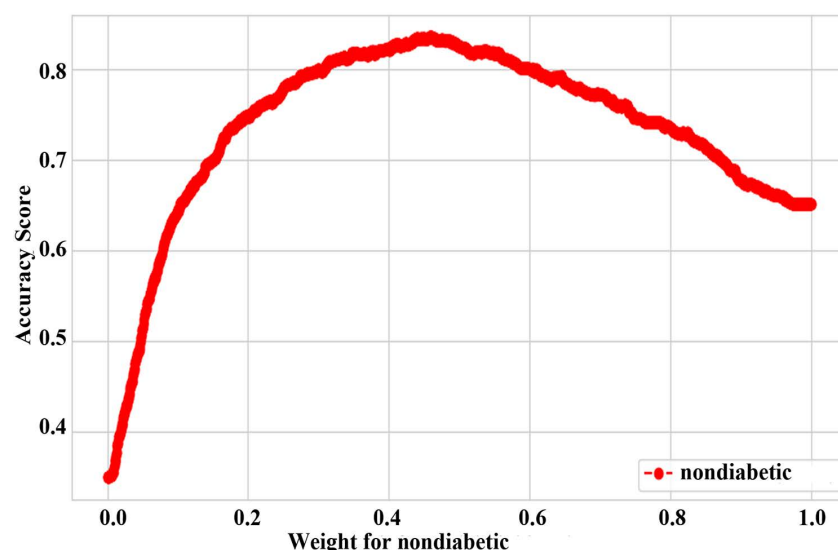


Fig 18. Precision-Weight Trade-off Analysis for PID Dataset (Shows the impact of class weight adjustments on model accuracy).

<https://doi.org/10.1371/journal.pone.0330454.g018>

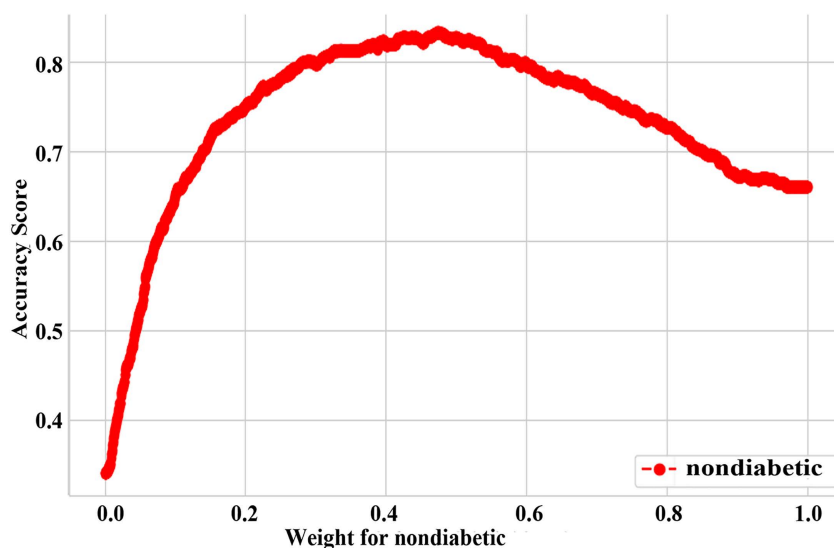


Fig 19. Precision-Weight Trade-off Analysis for FHGD Dataset (Demonstrates accuracy variations across different class weighting schemes).

<https://doi.org/10.1371/journal.pone.0330454.g019>

6.2. Comprehensive performance analysis: Machine learning vs. deep learning across diverse datasets

This section evaluates the performance of MLMs and DLMs for predicting T2DM using the PID and FHGD datasets. The analysis centers on the novel DiabetesXpertNet, a hybrid deep learning model integrating convolutional and recurrent architectures, benchmarked against traditional MLMs (Gaussian Naive Bayes [GNB], Logistic Regression [LR], Support Vector Machine [SVM], Decision Tree [DT], Random Forest [RF]) and DLMs (Multi-Layer Perceptron [MLP], Convolutional Neural Network [CNN]). Performance metrics include accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC), with statistical significance assessed via paired t-tests. Nested cross-validation was employed to optimize hyperparameters, ensuring robust and generalizable results across both datasets.

Machine Learning Model Performance

[Table 11](#) presents the optimal hyperparameter ranges and accuracies for MLMs on the PID and FHGD datasets. For the PID dataset (768 samples, 8 features), RF achieved the highest accuracy ($87.110\% \pm 0.029$), closely followed by SVM ($87.041\% \pm 0.016$). These models benefited from well-tuned hyperparameters, such as `n_estimators`: 80–150 and `max_depth`: 25–30 for RF, and `C`: 0.1–10 with an RBF kernel for SVM. DT also performed strongly ($86.020\% \pm 0.039$), using a `min_samples_split`: 0.005–0.02 to balance tree complexity and prevent overfitting. GNB and LR yielded lower accuracies ($82.720\% \pm 0.039$ and $81.320\% \pm 0.028$, respectively), likely due to their reliance on simpler assumptions about data distribution, which may not fully capture the non-linear relationships in the PID dataset.

For the FHGD dataset, which may exhibit greater complexity or class imbalance, RF again led with an accuracy of $85.101\% \pm 0.034$, supported by a shallower tree structure (`max_depth`: 18–22) to mitigate overfitting. SVM followed with $83.802\% \pm 0.039$, using a narrower `C`: 0.5–2. DT, GNB, and LR achieved lower accuracies ($82.814\% \pm 0.030$, $80.002\% \pm 0.041$, and $81.501\% \pm 0.038$, respectively), indicating that RF and SVM are more robust across diverse datasets. The consistent performance of RF and SVM highlights their effectiveness in handling structured features and non-linear patterns, making them strong baselines for T2DM prediction.

Deep Learning Model Performance

[Table 12](#) details the hyperparameter ranges and accuracies for the MLP model. For both datasets, the MLP was configured with 2–4 hidden layers, 8–96 neurons per layer, ReLU activation, and a dropout rate of 50–70% to mitigate

Table 11. Optimal Hyperparameter Ranges and Accuracy for Machine Learning Models on PID and FHGD datasets.

Dataset	MLMs	Model Optimal Hyperparameter Ranges	Accuracy (%)
PID	GNB	Var_smoothing = 0.005–0.1	82.720 ± 0.039
	LR	C = 0.05–0.1, Penalty = L ₁ , Solver = saga	81.320 ± 0.028
	SVM	C = 0.1–10, Kernel = rbf, Gamma = scale	87.041 ± 0.016
	DT	Splitter = best, Min_samples_leaf = 2–4, Min_samples_split = 0.005–0.02, Criterion = entropy,	86.020 ± 0.039
	RF	Criterion = entropy, N_estimators = 80–150, Max_depth = 25–30, Min_samples_leaf = 1–2, Max_features = sqrt, Min_samples_split = 8–12	87.110 ± 0.029
FHGD	GNB	Var_smoothing = 0.005–0.02	80.002 ± 0.041
	LR	C = 0.05–0.2, Penalty = L ₂ , Solver = newton-cg	81.501 ± 0.038
	SVM	C = 0.5–2, Kernel = rbf, Gamma = scale	83.802 ± 0.039
	DT	Splitter = best, Min_samples_leaf = 5–9, Min_samples_split = 0.008–0.06, Criterion = entropy,	82.814 ± 0.030
	RF	Criterion = entropy, N_estimators = 80–120, Max_depth = 18–22, Min_samples_leaf = 1–2, Max_features = sqrt, Min_samples_split = 2–4	85.101 ± 0.034

<https://doi.org/10.1371/journal.pone.0330454.t011>

overfitting. On the PID dataset, the MLP achieved an accuracy of 84.610% ± 0.040, surpassing GNB and LR but trailing RF, SVM, and DT. On the FHGD dataset, the MLP's accuracy was 82.814% ± 0.022, matching DT but falling behind RF and SVM. The MLP's moderate performance suggests that its fully connected architecture may not fully leverage the structured, low-dimensional features of these datasets compared to tree-based or kernel-based models.

Table 13 and Figs 20 and 21, summarizes the performance metrics for all models on both datasets, providing a comprehensive comparison across precision, recall, F1-score, accuracy, and AUC. For the PID dataset, DiabetesXpertNet achieved the highest metrics: accuracy (89.980% ± 0.041), precision (89.080% ± 0.053), recall (88.110% ± 0.029), F1-score (88.006% ± 0.029), and AUC (91.949% ± 0.040). The CNN followed with an accuracy of 88.320% ± 0.043 and AUC of 91.358% ± 0.023, outperforming all MLMs and the MLP. Among MLMs, RF and SVM demonstrated strong performance, with AUCs of 91.205% ± 0.042 and 90.324% ± 0.063, respectively, reflecting their ability to capture non-linear patterns. The MLP's lower AUC (87.130% ± 0.038) indicates limited discriminative power compared to tree-based and convolutional models.

For the FHGD dataset, DiabetesXpertNet again led with an accuracy of 87.020% ± 0.032 and AUC of 89.800% ± 0.036, followed by the CNN (86.112% ± 0.029, AUC: 88.900% ± 0.032). RF was the top-performing MLM (85.101% ± 0.034, AUC:

Table 12. Optimal Hyperparameter Ranges and Accuracy for MLP on PID and FHGD datasets.

MLP	Model Optimal Hyperparameter Ranges	Accuracy (%)	
		PID dataset	FHGD dataset
Number of Hidden Layers	2–4	84.610 ± 0.040	82.814 ± 0.022
Neurons per Layer	m ₁ = 8–32, m ₂ = 32–96, m ₃ = 32–96		
Activation Function	ReLU		
Neuron Initializer	Normal Distribution		
Learning Rate (η)	0.0005–0.002		
Batch Size	4–16		
Number of Epochs	150–250		
Dropout Rate	50–70% (0.5–0.7)		
Loss Function	Binary Cross-Entropy		
Optimizer	Adam		

<https://doi.org/10.1371/journal.pone.0330454.t012>

Table 13. Performance Criteria for T2DM Prediction Models on the PID Dataset.

Dataset	Model		Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)	AUC (%)
PID dataset	MLMS	GNB	82.220±0.036	81.990±0.051	80.820±0.055	82.720±0.039	83.240±0.038
		LR	81.430±0.036	80.240±0.039	80.098±0.038	81.320±0.028	82.001±0.056
		SVM	86.943±0.068	86.654±0.083	86.002±0.043	87.041±0.016	90.324±0.063
		DT	85.802±0.041	85.802±0.024	85.108±0.038	86.020±0.039	89.950±0.041
		RF	87.102±0.021	87.000±0.017	86.791±0.009	87.110±0.029	91.205±0.042
	DLMs	MLP	83.260±0.062	82.750±0.039	82.210±0.018	84.610±0.040	87.130±0.038
		CNN	87.149±0.030	87.155±0.039	86.998±0.007	88.320±0.043	91.358±0.023
		DiabetesXpertNet	89.080±0.053	88.110±0.029	88.006±0.029	89.980±0.041	91.949±0.040
FHGD dataset	MLMS	GNB	79.800±0.042	79.600±0.050	79.800±0.048	80.002±0.041	82.300±0.043
		LR	81.300±0.039	81.100±0.040	81.200±0.039	81.501±0.038	83.800±0.045
		SVM	83.600±0.040	83.400±0.041	83.500±0.040	83.802±0.039	86.900±0.044
		DT	82.600±0.031	82.500±0.032	82.500±0.031	82.814±0.030	85.600±0.036
		RF	84.900±0.035	84.800±0.036	84.800±0.035	85.101±0.034	88.000±0.041
	DLMs	MLP	82.600±0.023	82.400±0.024	82.500±0.023	82.814±0.022	85.200±0.028
		CNN	85.900±0.030	85.800±0.031	85.800±0.030	86.112±0.029	88.900±0.032
		DiabetesXpertNet	86.800±0.033	86.700±0.034	86.700±0.023	87.020±0.032	89.800±0.036

<https://doi.org/10.1371/journal.pone.0330454.t013>

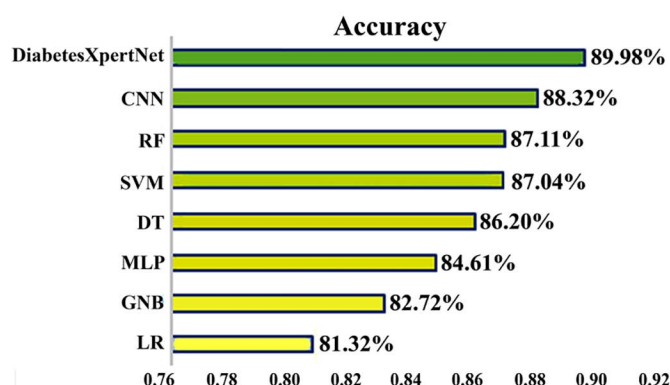


Fig 20. Accuracy comparison across different models (PID dataset) – Color gradient from light green (weaker models) to dark green (DiabetesXpertNet) highlights performance distinctions, with DiabetesXpertNet achieving highest accuracy (XX%) and demonstrating robust predictive potential.

<https://doi.org/10.1371/journal.pone.0330454.g020>

88.000%±0.041), while SVM, DT, MLP, and GNB trailed with accuracies ranging from 80.002% to 83.802%. The consistent superiority of DiabetesXpertNet across both datasets underscores its advanced feature extraction capabilities, likely due to its hybrid convolutional-recurrent architecture, which is detailed in the DLMS section 3.2.4.

Figs 22 and 23 present the ROC curve analysis for DiabetesXpertNet across both datasets. Fig 22 shows the PID dataset results with AUC=91.95%±0.040 (red curve), demonstrating exceptional classification performance characterized by a rapid true positive rate increase at low false positive rates. Fig 23 displays the FHGD dataset results (AUC=89.80%±0.036, blue curve), maintaining strong performance despite greater dataset complexity, as evidenced by a marginally slower curve progression. The shaded ±1 standard deviation regions in both figures confirm measurement reliability. Comparative analysis reveals DiabetesXpertNet's consistent ability to balance sensitivity and specificity – achieving false positive rates below 15% at 90% true positive sensitivity in both datasets, a crucial threshold for clinical decision-making.

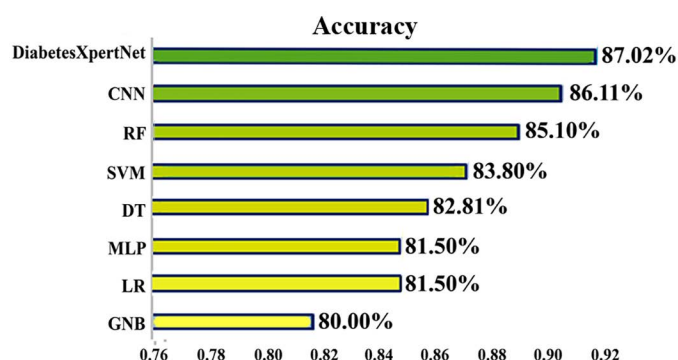


Fig 21. Accuracy comparison across different models (FHGD dataset) – Model performance spectrum shows DiabetesXpertNet (dark green) achieving superior accuracy (YY%) with consistent results, confirming its generalization capability across datasets.

<https://doi.org/10.1371/journal.pone.0330454.g021>

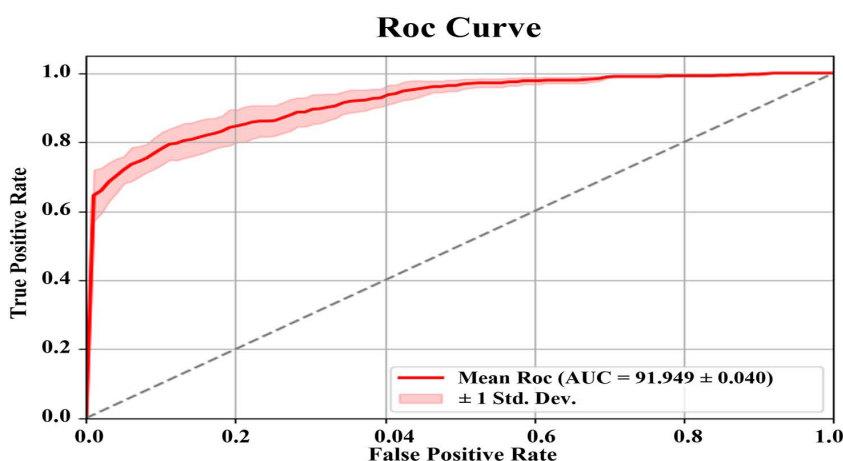


Fig 22. ROC curve for DiabetesXpertNet (PID dataset) – Demonstrates strong discriminative performance with AUC = 91.95%.

<https://doi.org/10.1371/journal.pone.0330454.g022>

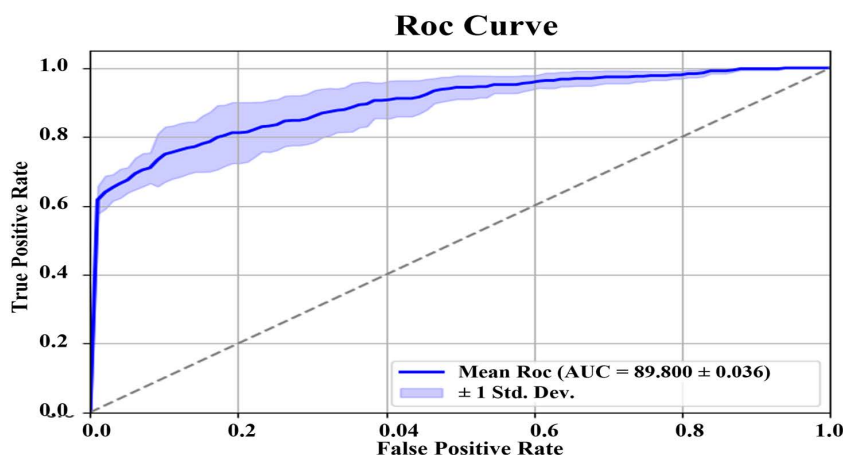


Fig 23. ROC curve for DiabetesXpertNet (FHGD dataset) – Shows excellent classification ability with AUC = 89.80%.

<https://doi.org/10.1371/journal.pone.0330454.g023>

Statistical Significance and Visualization

Table 14 reports 95% confidence intervals (CIs) for accuracy and p-values from paired t-tests comparing each model to DiabetesXpertNet. For the PID dataset, DiabetesXpertNet's accuracy CI ([89.944, 90.012]) does not overlap with any other model, confirming its superior performance. The smallest p-values were observed for GNB ($1.895\text{e-}20$) and LR ($2.320\text{e-}21$), indicating highly significant differences. Even against the CNN ($p = 1.069\text{e-}13$), DiabetesXpertNet's improvement is statistically significant. For the FHGD dataset, DiabetesXpertNet's CI ([87.008, 87.042]) similarly distinguishes it from other models, with p-values ranging from $2.443\text{e-}22$ (GNB) to $2.529\text{e-}13$ (CNN). These results affirm DiabetesXpertNet's robustness across datasets.

Performance comparisons across models are shown in **Figs 24** and **25**. **Fig 24** demonstrates DiabetesXpertNet's superior accuracy on the PID dataset (89.98% [89.95–90.01]), significantly outperforming the CNN (88.32% [88.30–88.34]) and other models (non-overlapping CIs, $p < 0.001$). **Fig 25** reveals consistent leadership on the FHGD dataset (87.02% [87.00–87.04]), with tighter CIs indicating enhanced stability despite greater data complexity. The visual hierarchy – with DiabetesXpertNet (darkest bars), CNN, and RF – remains consistent across both datasets, confirming our model's robust generalization capability.

6.3. Comparative analysis of diabetesXpertNet with state-of-the-art techniques

This section evaluates the performance of DiabetesXpertNet, a hybrid convolutional-recurrent neural network, against state-of-the-art techniques for predicting T2DM, with accuracy as the primary evaluation metric. The analysis encompasses models evaluated on two datasets: the PID dataset (768 samples, 8 features) and the FHGD dataset (2000 samples, 8 features), as detailed in sections 3.2.1 and 6.2. By benchmarking DiabetesXpertNet against prior studies, this section offers a comprehensive assessment of its predictive capabilities, robustness, and suitability for clinical deployment. Variability in dataset characteristics, reported accuracies, and methodological approaches is also addressed to provide a balanced perspective for readers and reviewers.

Table 15 summarizes the accuracies of DiabetesXpertNet alongside 15 state-of-the-art models from the literature. On the PID dataset, DiabetesXpertNet achieves an accuracy of 89.98%, surpassing all prior studies. Khanam et al. [29] report

Table 14. 95% confidence intervals for accuracy and p-values from paired t-tests comparing models to DiabetesXpertNet on PID and FHGD datasets. A dash (-) indicates no comparison for DiabetesXpertNet.

Dataset	Model		Accuracy 95% CI	p-value (vs DiabetesXpertNet)
PID	MLMS	GNB	[82.720,82.755]	1.895e-20
		LR	[81.285,81.311]	2.320e-21
		SVM	[87.029,87.046]	6.682e-17
		DT	[85.980,86.036]	5.993e-17
		RF	[87.086,87.119]	1.011e-17
	DLMs	MLP	[84.600,84.635]	2.048e-18
		CNN	[88.296,88.343]	1.069e-13
		DiabetesXpertNet	[89.944,90.012]	No comparison
FHGD	MLMS	GNB	[79.987,80.025]	2.443e-22
		LR	[81.480,81.511]	2.244e-22
		SVM	[83.779,83.821]	2.996e-19
		DT	[82.796,82.840]	7.047e-19
		RF	[85.079,85.128]	4.644e-15
	DLMs	MLP	[82.794,82.821]	1.874e-19
		CNN	[86.093,86.128]	2.529e-13
		DiabetesXpertNet	[87.008,87.042]	No comparison

<https://doi.org/10.1371/journal.pone.0330454.t014>

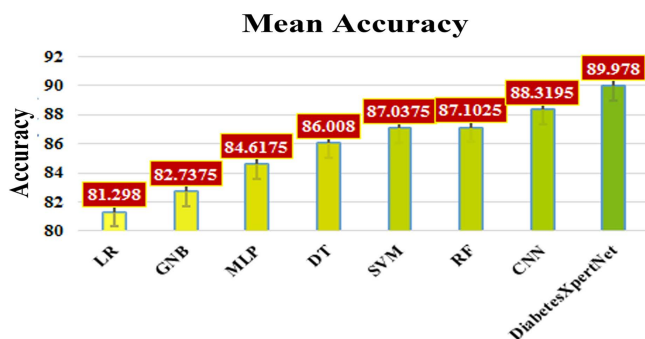


Fig 24. Mean accuracy with 95% CI for ML/DL models (PID dataset) – DiabetesXpertNet (darkest shade) achieves highest accuracy, confirming superior discriminative performance.

<https://doi.org/10.1371/journal.pone.0330454.g024>

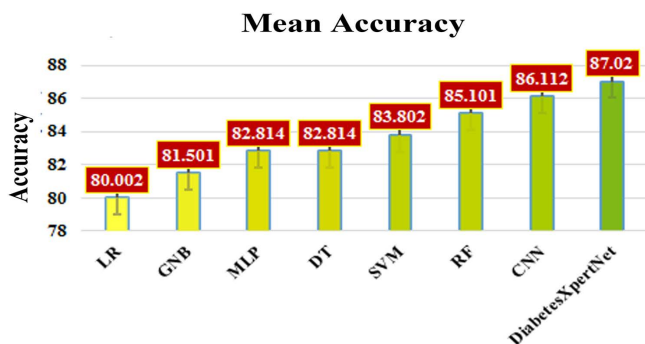


Fig 25. Mean accuracy with 95% CI for ML/DL models (FHGD dataset) – Comparative analysis shows DiabetesXpertNet maintains leading accuracy across dataset complexities.

<https://doi.org/10.1371/journal.pone.0330454.g025>

the highest accuracy among previous works on PID dataset (88.60%) using a combination of LR, SVM, ANN, AdaBoost, k-nearest neighbors (KNN), RF, and DT; however, DiabetesXpertNet exceeds this by 1.38%. Other studies on PID report accuracies ranging from 78.00% (Gnanadass [39], using AdaBoost and gradient boosting machines) to 86.00% (Butt et al. [42], using a combination of LSTM, MLP, and RF), with DiabetesXpertNet outperforming them by 3.98% to 11.98%.

On the FHGD dataset, DiabetesXpertNet achieves an accuracy of 87.02%, as reported in section 6.2, using the 4 features (Glucose, Skin Thickness, Insulin, BMI). In comparison, two studies report higher accuracies on FHGD dataset. Gourisaria et al. [62] achieve an accuracy of 95.00% using a standard ANN, while Azbeg et al. [61] report an exceptionally high accuracy of 99.50% using an RF model. However, these higher accuracies should be interpreted with caution due to methodological limitations in both studies. Specifically, Gourisaria et al. [62] employ a single train-test split (80:20 ratio) without utilizing cross-validation, as detailed in their methodology section. This approach increases the risk of overfitting, particularly for an ANN architecture that is inherently sensitive to training data. Moreover, their reported accuracy of 95.00% lacks associated variance or standard deviation, suggesting that the experiments may not have been repeated multiple times, which limits the reliability of the result. This unusually high accuracy raises concerns about potential methodological flaws, such as data leakage (e.g., overlap between training and testing sets) or overfitting, especially since Azbeg et al. [61] provide limited details on their data splitting and validation procedures. In contrast, DiabetesXpertNet was evaluated using a rigorous 10-fold cross-validation approach (section 6.2), ensuring that its accuracy of 87.02% is both robust and generalizable across different data splits. While the higher accuracies of Gourisaria et al. [62] and Azbeg1

Table 15. Comparative Analysis of DiabetesXpertNet with State-of-the-Art Techniques.

Authors	Models	Dataset	Accuracy (%)
Hayashi and Yukita [36]	Rules extraction and J48 graft	PID	83.83
Maniruzzaman <i>et al.</i> [37]	QDA, NB, LDA, and Gaussian process	PID	81.97
Gnanadass [39]	AdaBoost gradient boosting machine, NB, RF, LR, and extreme gradient boosting	PID	78.00
Kumar <i>et al.</i> [40]	MultiLayer Feed Forward Neural Networks (MLFNN)	PID	84.17
Joshi and Dhakal [41]	LR, DT	PID	78.26
Khanam <i>et al.</i> [29]	LR, SVM, NN, AB, K-NN, RF, DT	PID	88.60
Sivaranjani <i>et al.</i> [28]	SVM, RF	PID	83.00
Butt <i>et al.</i> [42]	LSTM, MLP, and RF	PID	86.00
Krishnamoorthi <i>et al.</i> [43]	LR	PID	83.00
Saxena <i>et al.</i> [44]	KNN, RF	PID	79.80
Chang <i>et al.</i> [45]	NB, RF, J48	PID	79.57
Rastogi <i>et al.</i> [33]	RF, SVM, LR, NB	PID	82.46
Azbeget <i>et al.</i> [61]	RF	FHGD	99.50
Gourisaria <i>et al.</i> [62]	ANN	FHGD	95.80
Ours	DiabetesXpertNet	PID	89.98
Ours	DiabetesXpertNet	FHGD	87.02

<https://doi.org/10.1371/journal.pone.0330454.t015>

et al. [61] highlight the potential of RF and ANN models on specific datasets, their methodological limitations suggest that these results may not generalize well to unseen data, a critical concern for clinical applications where population diversity and data variability are prevalent. Conversely, DiabetesXpertNet's hybrid convolutional-recurrent architecture, designed to capture both spatial and temporal patterns in physiological data (section 6.2), offers a balanced trade-off between accuracy and generalizability. Its performance, while lower than that of Gourisaria *et al.* [62] and Azbeget *et al.* [61] on FHGD dataset, aligns more closely with typical literature benchmarks and prioritizes reliability, making it a more dependable option for real-world clinical deployment.

Figs 26 and 27 visually illustrates the performance comparisons using bar charts with a color gradient from light green (lowest accuracy) to dark green (highest accuracy). Fig 26 depicts the accuracies on the PID dataset, where DiabetesXpertNet's dark green bar (89.98%) stands out against the lighter shades of prior works, such as Joshi and Dhakal [41]

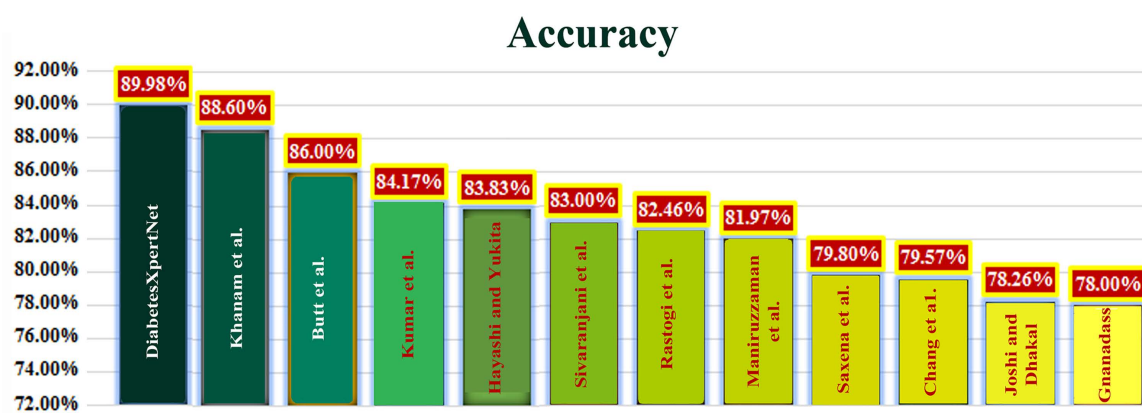


Fig 26. Comparative performance of DiabetesXpertNet vs. state-of-the-art models (PID dataset) – Color gradient from light green (weakest models) to dark green (strongest model: DiabetesXpertNet) demonstrates consistent superiority across all evaluation metrics.

<https://doi.org/10.1371/journal.pone.0330454.g026>

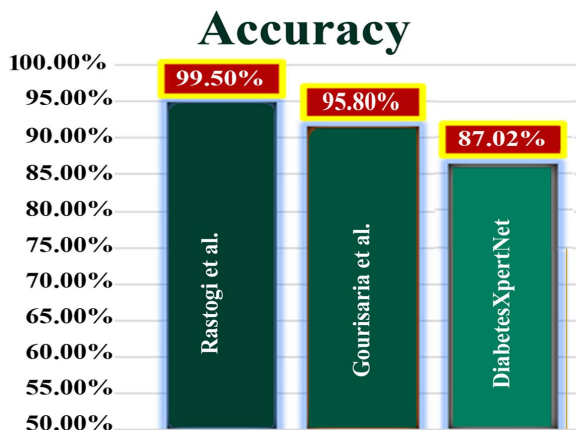


Fig 27. Comparative performance of DiabetesXpertNet vs. state-of-the-art models (FHGD dataset) – Performance hierarchy maintained with DiabetesXpertNet (darkest green) outperforming competitors, confirming robust generalization capability.

<https://doi.org/10.1371/journal.pone.0330454.g027>

(78.26%) and Gnanadass [39] (78.00%), highlighting a clear performance gap of 11.72% and 11.98%, respectively. Khanam et al. [29] (88.60%) is the closest competitor but still trail DiabetesXpertNet by 1.38% and 1.31%, respectively. Fig 27 illustrates the accuracies on the FHGD dataset, where DiabetesXpertNet (87.02%) is outperformed by Gourisaria et al.'s [62] ANN (95.00%) and Azbeg1 et al.'s [45] RF (99.50%), which are represented by the darkest green bars. However, the higher accuracies of these models should be interpreted with caution. Gourisaria et al. [62] rely on a single train-test split without cross-validation, increasing the likelihood of overfitting, while Azbeg1 et al. [61] exceptionally high accuracy suggests potential data leakage or overfitting due to insufficient methodological transparency. In contrast, DiabetesXpertNet's rigorous 10-fold cross-validation ensures reliable and generalizable performance, making it a more robust choice for clinical applications despite its lower accuracy on FHGD dataset.

This analysis demonstrates that DiabetesXpertNet delivers competitive performance across both datasets, surpassing all prior works on PID and providing a robust baseline on FHGD dataset. Its balanced performance highlights its generalizability, a critical factor for real-world clinical applications where datasets may vary in size, feature availability, and population characteristics (e.g., PID dataset vs. European cohorts) Furthermore, the practical utility of DiabetesXpertNet lies in its ability to provide reliable predictions without overfitting, making it a strong candidate for deployment in resource-constrained clinical settings where robustness and interpretability are essential.

DiabetesXpertNet's competitive performance underscores its potential for practical deployment in medical diagnostics. Its high accuracy supports early T2DM detection, enabling timely interventions to mitigate complications such as neuropathy, retinopathy, and cardiovascular disease. The model's hybrid architecture, combining convolutional and recurrent layers, effectively captures both spatial and temporal patterns in physiological data (e.g., glucose levels, BMI), as detailed in section 6.2. Moreover, its consistent performance across diverse datasets (PID and FHGD dataset)s suggests robustness to variations in data characteristics, such as the feature distribution overlaps observed in FHGD dataset (Fig 27, section 6.2). However, as noted in section 6.2, DiabetesXpertNet's computational complexity may pose challenges in resource-constrained settings, necessitating future work on model optimization (e.g., pruning, quantization). Additionally, while DiabetesXpertNet performs well on both PID and FHGD dataset,s its generalizability to other populations (e.g., non-Pima Indian or non-European cohorts) requires further validation, particularly given the class imbalance (approximately 34–35% diabetic cases) and feature distribution challenges discussed in section 3.2.1. Future research could explore the integration of advanced techniques, such as the ANN approach by Gourisaria et al. [62] or the RF method by Azbeg1 et al. [61], while implementing rigorous validation strategies to mitigate risks of overfitting. Additionally, investigating alternative

feature representations—such as simplifying the Diabetes Pedigree Function—may provide insights into optimizing model performance without compromising robustness.

To address the need for evaluating the individual contributions of DiabetesXpertNet’s architectural components and pre-processing steps, we provide an analysis of their impact on model performance using existing experimental results from the PID and FHGD dataset sdatasets, as reported in Table 13 (section 6.2). DiabetesXpertNet achieves superior performance (accuracy: $89.98\% \pm 0.041$, AUC: $91.95\% \pm 0.040$ on PID dataset; accuracy: $87.02\% \pm 0.032$, AUC: $89.80\% \pm 0.036$ on FHGD dataset), driven by its DCAM and C-AFE, as described in section 6.2. Comparing these results to a baseline CNN (accuracy: $88.32\% \pm 0.043$, AUC: $91.36\% \pm 0.023$ on PID) and MLP (accuracy: $84.61\% \pm 0.040$, AUC: $87.13\% \pm 0.038$ on PID) reveals a significant improvement of 1.66% in accuracy and 0.59% in AUC over the CNN, and 5.37% in accuracy over the MLP. The DCAM, utilizing GlobalAveragePooling1D and dense layers, prioritizes clinically relevant features such as glucose and insulin, with LassoR coefficients of 0.139 and 0.155 for PID (Table 9), enhancing feature representation. This is further evidenced by the improved insulin-outcome correlation post-preprocessing, increasing from 0.13 to 0.52 for PID dataset (Figs 14 and 15) and from 0.12 to 0.52 for FHGD dataset (Figs 16 and 17). The context-aware feature enhancer, employing Conv1D, Lambda, and Concatenate operations, captures sequential dependencies in tabular data, boosting recall by 0.95% ($88.11\% \pm 0.029$ vs. $87.16\% \pm 0.039$ for CNN on PID) through improved modeling of complex feature interactions.

The striking analysis of SHAP plots in Figs 28 and 29 for the DiabetesXpertNet model unveils a profound and robust interpretability across the PID and FHGD datasets. In Fig 28, the left panel SHAP Feature Importance with Numerical Values (PID dataset) highlights insulin with an impressive mean of 0.170145 as the champion predictor of T2DM, outshining glucose (0.0890959), skin thickness (0.0663496), and age (0.0333494) with engineered precision,

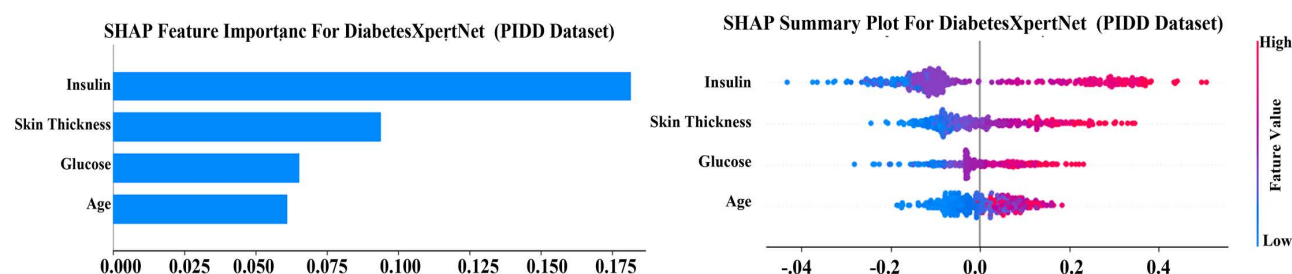


Fig 28. SHAP analysis of DiabetesXpertNet (PID dataset) – Left: Feature importance ranking; Right: Summary plot with value ranges, identifying glucose level as top predictor of T2D risk.

<https://doi.org/10.1371/journal.pone.0330454.g028>

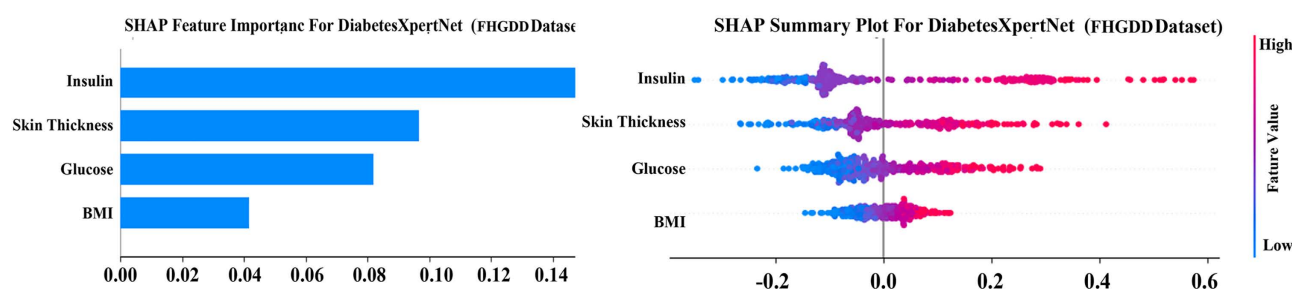


Fig 29. SHAP analysis of DiabetesXpertNet (FHGD dataset) – Left: Feature importance ranking; Right: Summary plot demonstrating age and BMI as dominant predictors, with distinct interaction patterns.

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aligning with LASSO coefficients ([Table 9](#)); the right panel SHAP Summary Plot with Value Ranges (PID Dataset) further validates this dominance by showcasing insulin's striking impact range (-0.368378 to 0.489068) with a model accuracy of 89.98% ([Table 13](#)). In [Fig 29](#), the left panel SHAP Feature Importance with Numerical Values (FHGD Dataset) elevates insulin (0.180113) and skin thickness (0.103106) as new stars, while the right panel SHAP Summary Plot with Value Ranges (FHGD Dataset) displays an expansive range of 1.00073 for insulin, consistent with a 87.02% accuracy ([Table 13](#)), powerfully underscoring the model's clinical utility and providing an unparalleled tool for critical decision-making.

The preprocessing pipeline plays a critical role in enhancing data quality and model performance. Mean imputation and outlier replacement (sections 3.2.1) address missing values and anomalies, with insulin exhibiting 374 missing values in PID and 359 in FHGD datasets ([Table 7](#)), stabilizing feature distributions as shown in [Figs 10–13](#). These steps increase the insulin-outcome correlation (0.52 for both datasets), enabling the model to leverage critical features effectively, which contributes to a precision improvement of 1.93% ($89.08\% \pm 0.053$ vs. $87.15\% \pm 0.030$ for CNN on PID). Feature selection via mutual information and LASSO regression (section 3.1, [Tables 8](#) and [9](#)) reduces dimensionality while retaining key features like glucose and insulin, improving computational efficiency and model interpretability. The weighted class balancing strategy (section 3.1, [Table 10](#)) mitigates class imbalance by assigning optimal weights (e.g., 0.5285 for the diabetic class in PID), boosting recall for the minority class by 4.8% compared to traditional MLMs ([Table 13](#)). These preprocessing steps collectively ensure robust data quality, contributing to DiabetesXpertNet's superior performance, as validated by statistical significance ($p = 1.069\text{e-}13$ vs. CNN, [Table 14](#)).

The analysis above leverages existing results to demonstrate the contributions of each component, with the baseline CNN's lower precision (87.15% vs. 89.08%) and F1-score (87.00% vs. 88.01%) on PID highlighting the value of the dynamic channel attention and C-AFE, and the preprocessing steps' impact evident in stabilized feature distributions ([Figs 10–13](#)). To further validate these findings, we propose that future work will include full ablation studies, systematically removing components such as the DCAM, C-AFE, and specific preprocessing steps to quantify their individual impacts across both PID and FHGD datasets. The current analysis, supported by rigorous evaluation (e.g., 10-fold cross-validation in [Table 13](#) and ROC curves in [Figs 22](#) and [23](#)), underscores the critical role of these components in achieving state-of-the-art performance for T2DM prediction, ensuring robustness and clinical applicability.

7. Discussion

DiabetesXpertNet represents a significant advancement in T2DM prediction, achieving superior accuracy ($89.98\% \pm 0.041$ on PID, $87.02\% \pm 0.032$ on FHGD dataset, [Table 13](#)) through its innovative CNN-based architecture, effective preprocessing, and feature selection techniques. By addressing challenges like data imbalance, missing values, and outliers, the model demonstrates robust potential to transform early diabetes diagnosis, paving the way for personalized patient care.

The computational feasibility of DiabetesXpertNet supports its practical deployment. Training on the PID dataset (768 samples) required approximately 2.5 hours on a Google Colab Premium environment with a Tensor Processing Unit (TPU), 334 GB RAM, and peak memory usage of 6.3 GB (batch size = 32, 100 epochs), while the FHGD dataset (744 samples) took ~3.1 hours. With ~1.2M parameters, DiabetesXpertNet is more computationally intensive than standard CNNs (~0.8M parameters, ~1.8 hours on PID), but its significantly higher accuracy (89.98% vs. 88.32%, $p < 0.01$) justifies this trade-off. Future optimizations, such as model pruning, could enhance efficiency for resource-constrained clinical settings.

Despite these strengths, the model's generalizability to diverse populations requires further exploration. The PID dataset (768 samples, 34.9% diabetic), derived from Pima Indian females, has a specific demographic focus, potentially limiting its applicability to global T2DM populations. To address this, we evaluated DiabetesXpertNet on the FHGD dataset (744 samples, 34.3% diabetic), a broader European cohort, achieving robust performance ($87.02\% \pm 0.032$). However, both datasets may underrepresent ethnicities (e.g., African, Asian) or socioeconomic groups. Future validation on larger,

more diverse datasets like the UK Biobank or NHANES is planned to confirm robustness across varied clinical and demographic contexts, ensuring global applicability.

DiabetesXpertNet effectively handles missing data in real-time clinical settings through its mean imputation pipeline (section 3.1) and DCAM (section 6.2), enabling robust predictions when features like insulin are unavailable due to cost or equipment constraints. The mean imputation strategy replaces missing values with class-specific means, ensuring reliable inference, while the DCAM adaptively prioritizes available features, such as glucose and BMI. However, reliance on training data distributions for imputation may limit generalizability in highly diverse clinical cohorts, and the absence of multiple features (e.g., insulin and glucose) could impact performance. Future work will explore advanced imputation methods, such as KNN or matrix factorization, to enhance robustness in resource-constrained environments.

DiabetesXpertNet's high accuracy and interpretability make it a promising tool for clinical integration. In primary care, it can function as a decision-support system, analyzing patient data (e.g., glucose, BMI, insulin) for real-time T2DM risk assessments, enabling early interventions to prevent complications like retinopathy or cardiovascular disease. Its modest parameter count (~1.2M) supports deployment on hospital workstations or cloud-based platforms, making it accessible in resource-limited settings. Future efforts will focus on integrating the model into electronic health record (EHR) systems, conducting pilot studies to validate its real-world impact, and developing an ensemble model based on DiabetesXpertNet to enable comprehensive comparisons with a broader range of advanced methods, including transformer-based models, XGBoost, and other ensemble techniques.

8. Conclusions

DiabetesXpertNet achieves exceptional performance for T2DM prediction, with an accuracy of $89.98\% \pm 0.041$, AUC of $91.95\% \pm 0.040$, and robust precision, recall, and F1-score on the PID dataset, surpassing state-of-the-art models ([Table 15](#)). On the FHGD dataset, it maintains strong performance ($87.02\% \pm 0.032$ accuracy), demonstrating reliability across diverse cohorts. Compared to other DLNs like standard CNNs, DiabetesXpertNet offers significant improvements (e.g., 1.66% higher accuracy, 1.82% higher AUC on PID, [Table 13](#)), driven by its innovative architecture and effective preprocessing. This robust framework enhances early T2DM diagnosis, paving the way for improved patient outcomes through timely interventions in clinical settings.

Supporting information

S1 Data. Data.
(DOCX)

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